Travelling Wave Dynamics in Housing Markets
A Proposed Endogenous Leading Indicator of Crashes

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Abstract

Housing market crashes cause substantial economic harm yet are commonly modelled as exogenous shocks. Patterns documented separately suggest otherwise: a spatiotemporal 'ripple effect', and an 18-year land value periodicity in heterodox analysis. Inspired by Ricardo's Law of Rent, we ask whether an ordinal structure underlies the propagation of relative growth.

Using transactional data for England and Wales, 1995–2026, covering two cycles and one crash, we document a travelling wave: relative growth leadership, independent of aggregate growth rates, begins in the highest-priced segments and propagates monotonically towards the lowest over multi-year cycles, robust across geographic and quantile representations.

We model the wave by an advection equation whose velocity follows from the log-logistic price structure. The cycle time is the informational distance in the land value gradient divided by the rate at which competitive reallocation processes it. As the wave reaches the margin, capital accumulates against the boundary; when that renders land values there unsustainable, a behavioural response triggers a correction. The wave's position is thus a leading indicator of the cycle.

The central claims, that the cycle is endogenous and ends at the margin, are offered as falsifiable hypotheses. If correct, the account informs the debate over land value taxation.

1 Introduction

1.1 Spatiotemporal Patterns

Prior work has documented the 'ripple effect'[2–4] across UK regions in which house price growth propagates from London outwards. Our approach extends and refines this literature in three respects. First, whilst we investigate geographic space, we adopt quantile space (Section 2), which avoids both within-geography price heterogeneity and arbitrary administrative borders. Second, we model propagation as a kinematic travelling wave (Section 3) of ordinal growth ranks with a geometric boundary condition, not as statistical correlation. Third, we predict that cycles terminate at the distributional boundary (Section 4), rather than merely documenting that they spread.

A recent econophysics treatment of French housing[5] documents diffusive structure in the geographic field of price levels, with logarithmic decay of spatial correlations consistent with a 2D random diffusion equation. Our work on the propagation of relative growth ordering through the cross-sectional price distribution is complementary.

1.2 18-year Periodicity

A strand of heterodox analysis has documented persistent periodicities in land values. Hoyt[6] identified peaks in Chicago land values at intervals of approximately 18-20 years, a pattern later characterised by Foldvary[7] as a regular 18-year cycle across the United States. Harrison[8] has also observed the same periodicity in UK house prices. The recurrence of a similar cycle across time and place, under very different macroeconomic conditions, suggests an underlying mechanism worth investigating.

Our framework yields a cycle length consistent with the geometry of the price distribution and the rate of competitive reallocation (Sections 3.6 and 3.7), thereby providing a rationale for a pattern that has otherwise been observed without explanation.

1.3 Ricardo's Law of Rent

The classical economists Adam Smith, Thomas Malthus, and David Ricardo wrote extensively about land and the theory of rent[9]. Ricardo developed his Law of Rent in Chapter 2, 'On Rent', in *On the Principles of Political Economy and Taxation*[10]:

> *When, in the progress of society, land of the second degree of fertility is taken into cultivation, rent immediately commences on that of the first quality, and the amount of that rent will depend on the difference in the quality of these two portions of land.*
>
> *When land of the third quality is taken into cultivation, rent immediately commences on the second, and it is regulated as before, by the difference in their productive powers.*

Ricardo's sequential extension of cultivation implies that rent on newly-introduced land transitions from zero to positive, while rent on superior land rises by a smaller proportion. The locus of maximum relative growth therefore propagates monotonically down the quality ordering, orthogonal to changes in absolute land values. To our knowledge, this ordinal propagation structure has not previously been formalised.

Land's locational value in today's economy is equivalent to land's productive value in Ricardo's economy. A Georgian house in Liverpool is not worth as much as an identical property in London, and this difference is not due to its rebuild cost but rather its land, or locational, value. Recent work[11] documents that 80 percent of the post-war rise in real house prices was due to the land component. Across all properties there exists a continuum, a land value gradient, from where land consists of the majority of the value, to the margin where the value of land is zero; we use 'margin' throughout in this classical sense.

We invoke the Law of Rent not as a premise to be assumed but as the lens through which we recognised the travelling wave pattern. From it we take three structural claims, supported by modern transactional data: the ordinal propagation of relative growth, the land value gradient down which it propagates, and the constraining role of the margin. The contributions (Section 1.4) that follow stand on their own. We note that Ricardo's extension of cultivation is secular and monotonic, driven by population growth over long horizons; our phenomenon is cyclical, traversing the distribution and resetting over multi-year periods.

1.4 Contributions

This paper offers four falsifiable claims, each of which can be tested against both partial and completed cycles.

First, an empirical regularity (Section 2): the ordered propagation of relative growth leadership through the price distribution. This travelling wave provides an endogenous multi-year leading indicator of the housing cycle based on its position.

Second, a phenomenological coordinate framework (Sections 3.2, 3.3, 3.4, 3.5): the log-odds transformation, grounded in the empirically verified log-logistic price structure, in which the nonlinear quantile-space dynamics reduce to uniform advection, and in which the Jacobian, the velocity factor, the logistic density, and the Fisher information all coincide.

Third, a measured propagation speed κ (Section 3.4), approximately constant within each observed cycle, which we interpret as the rate at which competitive reallocation processes the land value gradient. It governs the growth field and, under the given representation, the capital density (Section 3.6).

Fourth, a boundary condition (Section 4) separating what the framework derives from what it assumes: the kinematic structure derives the accumulation of capital density at the margin as the wave arrives, while the price correction requires a minimal behavioural premise, that lenders and buyers respond to unsustainable land values at the margin.

2 Empirical Results: The Travelling Wave

2.1 Ordered Propagation of Relative Growth

In Figure 1, we combine a graph of the absolute smoothed log returns by decile with a heat map of the relative growth ranks of the same. A price per square metre dataset provides the best available proxy for land value (Section A.1).

The growth rank heat map visualises the relative ordering of price growth across deciles, independently of the growth levels. The fastest growing (or slowest falling) areas are in dark red, and the slowest growing (or fastest falling) areas are in dark blue.

Four representations, across 317 quantiles and Local Authorities, and price and price per square metre datasets (Figures A13 and A14), plus the decile plots of the same in smoothed and un-smoothed versions (Figures A16 and A17), confirm that the pattern is not an artefact of aggregation or smoothing.

Despite differing rates of log return across two housing cycles, we observe a monotonic propagation of relative growth leadership through the price distribution. The specific object that translates is a ridge in the two-dimensional growth field: at any given time, relative growth forms a peak at a particular quantile, with progressively lower growth on either side. We use *ridge* for the profile of the growth field at a given time, *wave peak* for the maximum of the ridge at any given time, and *travelling wave* or *wave* for its propagation as a whole. We derive the logistic trajectory for the wave peak from the log-logistic price structure (Section 3.5).

The apparent truncation in 2006 is a superposition artefact; the initial wave did in fact reach the margin of the entire distribution in the lowest-priced regions immediately before the crash (Section 2.2).

In the second cycle from 2010, the travelling wave starts again before stalling during the austerity years. When it does continue, it is parallel to

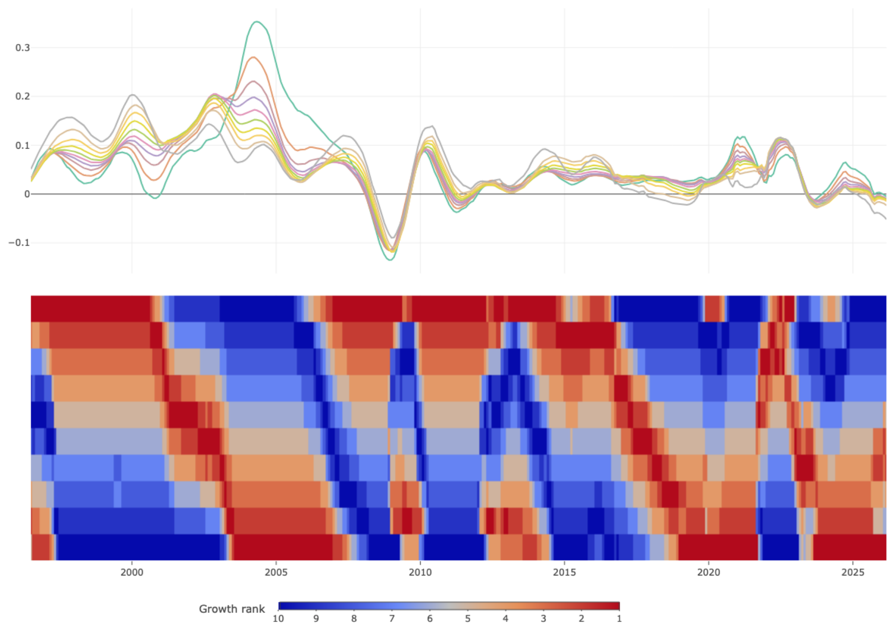

Figure 1: Top: Annual log return of the price per square metre dataset, by decile. Bottom: Growth rank of the same, by decile; highest decile at the top, lowest decile at the bottom. The logistic trajectory $q^*(t)$ (where q^* is the fastest-growing quantile each month, Section 3.5), of the wave peak in dark red, from upper to lower deciles over time, is visible in both cycles, though truncated in the first by the superposition of a second wave, and interrupted in the second by the 'race for space'.

the pattern from the earlier cycle. In 2022-23, there was a deviation from the existing ordering: the fastest growth was in larger, higher-priced properties. This was due to the 'race for space',[12;13] with buyers preferring more space following various COVID-19 lockdowns. After this period, the travelling wave peak resumed at the lowest-priced decile and has remained there; the interruption is seen particularly clearly in the z-space representation (Figure A20). The austerity and pandemic pattern of suspension and resumption suggests that the wave tracks a cumulative process that can be interrupted but not reset by transient shocks.

Figure 2 shows the wave peak trajectory of Figure 1 in log-odds space, before, during, and after the active traversal, from highest to lowest deciles. We plot 6,065 fitted trajectories from a grid search of 8,125 windows; 75% of all windows have $R^2 \geq 0.8$; the passing trajectories cluster tightly, so the slope is well-determined where the linear fit is good. Each line represents a linear fit over a different window (cycle start ± 12 months, length ± 6 months, matched across cycles). Each trajectory must include both the top and bottom decile. The concentration of lines shows the slope estimate is not an artefact of any single window choice. Since the windows overlap substantially the spread understates true sampling uncertainty, it is a window-sensitivity range rather than a confidence interval.

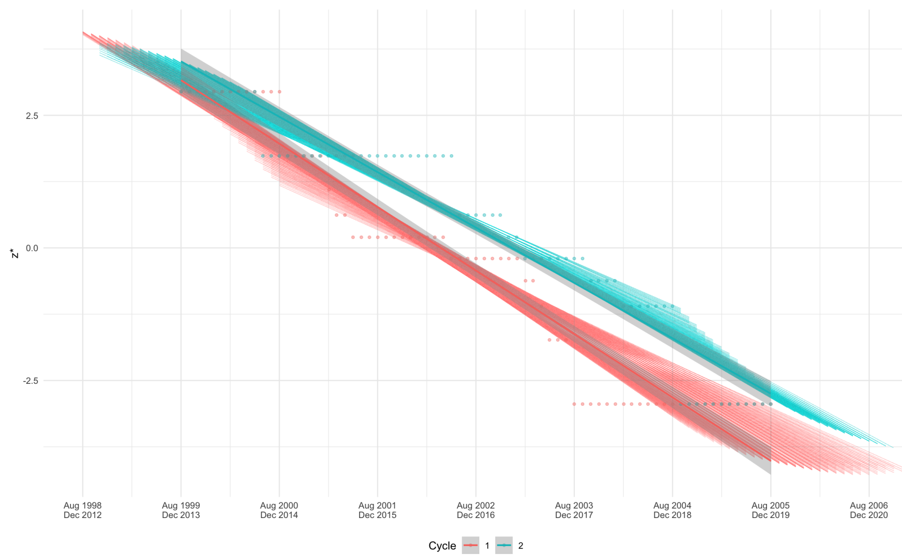

Figure 2: Wave peak trajectories in log-odds space, $z^* = \log(q^*/(1-q^*))$. This coordinate is the natural metric for the logistic price structure (Section 3.2); linear trajectories indicate the wave traverses the distribution at constant speed κ (Section 3.4). Trajectories are fitted by a grid search over windows to show that the slope is not an artefact of any single window choice. Median $\kappa_1 = 1.17$ $[1.08 - 1.21]$ and $\kappa_2 = 1.03$ $[0.97 - 1.06]$; the IQRs are window-sensitivity ranges, not confidence intervals, as the windows overlap.

2.2 Regional Analysis

In Figure 3, we provide a decomposition of Figure 1 by a subset of regions, up to 2010. All regions over the entire time period are shown in Figure A18.

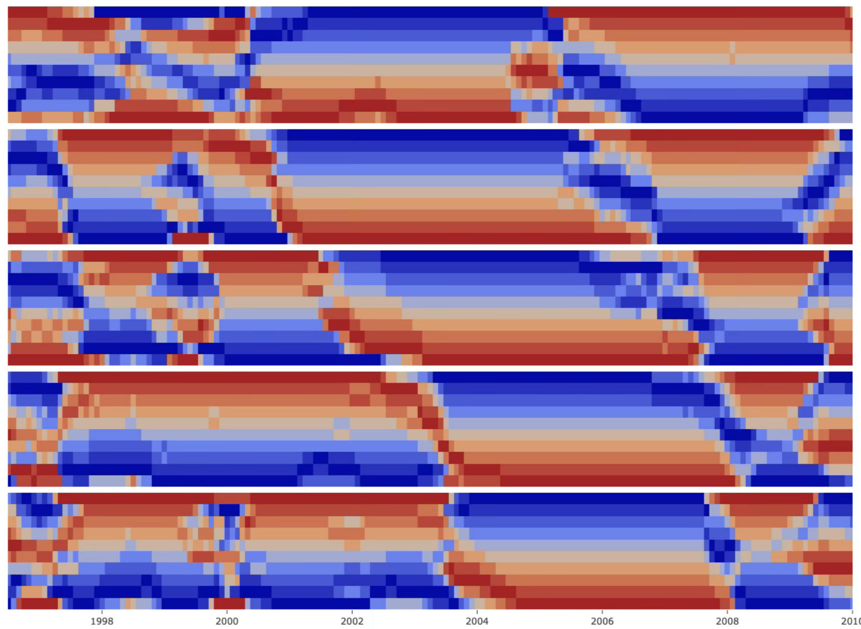

Figure 3: Growth rank of the annual log return of the monthly median, by decile; first cycle to 2010 only; selected regions arranged by descending median price: London, South East, East Midlands, North West, North East. The logistic trajectory of the wave peak from upper to lower deciles is visible to varying degrees in all plots.

The travelling wave appears within each region independently, not just at the national level. As the regions are ordered from top to bottom by median price, there is a cascade of travelling waves across regions: waves within an overall wave.

Second waves started in each region at different times, explaining the aggregate superposition in Figure 1. However, the initial wave for the North West, which contains some of the lowest-priced Local Authorities nationally, was still propagating until 2008, immediately prior to the crash.

The travelling wave also appears independently by property type (Figure A19).

3 Travelling Wave Model

The framework is phenomenological and operates at three layers, in increasing order of inference.

Empirically (Section 2), relative growth leadership propagates monotonically through the log-logistic price distribution at an approximately constant rate, κ, within each cycle (Figure 2).

Mathematically, we represent these observations as the kinematic advection of a growth ridge. Section 3.2 provides the evidence for the logistic distribution and introduces the standardised log-price coordinate z. Section 3.3 decomposes the growth field into an aggregate drift, a reshaping term, and the ridge, and identifies the ridge with the travelling wave. Section 3.4 models the uniform advection of the ridge with speed κ. Section 3.5 maps to quantile space, where the observable logistic trajectory emerges. Section 3.6 extends the same velocity field to the capital density. Section 3.7 yields the cycle time $T = \Delta z/\kappa$, a consistency result, from the geometry of the price distribution and speed κ.

Economically, Section 3.6 argues that competitive reallocation of capital (predominantly mortgage credit) is a candidate driver of the propagation. Section 4 establishes the boundary arrival of the wave as the endogenous condition for the cycle's terminal phase. In doing so, it derives the configuration but assumes a behavioural economic trigger: unsustainable land values at the margin; it rests on a single completed cycle.

The structure follows Lighthill and Whitham[14], who introduced kinematic waves for traffic flow with conserved density, a phenomenological flow-density relation, and shock formation when smooth advection breaks down. We adopt this form with two differences: the velocity field follows from the log-logistic structure, rather than being fitted as a flow-density relation, and instability arises at the distributional boundary rather than from interior shocks. The object being advected differs from the Fisher-KPP[15;16] class, which describe the propagation of a density or a frequency. Our wave is a ridge in the growth field, advected through the host distribution which it reshapes as it passes (Section 3.2). We are not aware of prior work treating travelling wave propagation through cross-sectional price distributions.

3.1 Variables

In Table 1, we define the symbols and units used throughout the analysis.

Table 1: Symbols, definitions, and units.

Symbol	**Definition**	**Unit**
P	The house price distribution	Price
$q \in (0, 1)$	The quantiles of the house price distribution	-
$\ell(q, t) = \log P(q, t)$	The log-price distribution, with location $\mu(t)$ and scale $s(t)$	Log-price
$z = \frac{\ell - \mu}{s} = \log\left(\frac{q}{1-q}\right)$	Standardised log-price; the log-odds of the quantile	Log-odds
$g(q, t) = \partial_t \ell(q, t)$	The growth field	Year^{-1}
$\bar{g}(t)$	The capital-weighted mean of the growth field	Year^{-1}
$r(q, t)$	The growth ridge	Year^{-1}
$f(q, t)$	The capital density with $\int f(q, t)dq = 1$	-
$q^*(t) = \arg\max_q g(q, t)$	The wave peak in q-space	-
$z^* = \log\left(\frac{q^*}{1-q^*}\right)$	The log-odds position of the wave peak	-
κ	The traversal rate: the approximately constant speed of the wave peak in z; measured as the magnitude of the slope of $z^*(t)$ and defined per cycle.	Log-odds year^{-1}

3.2 Logistic Log-Price Structure

Empirically, log-prices follow a logistic relationship across quantiles (Figure 4),

$$\ell(q,t) = \mu(t) + s(t)\log\left(\frac{q}{1-q}\right),$$

with a minimum R^2 of 0.987 (price) and 0.973 (price per square metre); the Cullen–Frey coordinates (Figure A4) place the distribution in the logistic/log-logistic region, with kurtosis near 4.2, distinct from the Gaussian value of 3. Each unit step in log-odds corresponds to the same increment in log-price. The scale or slope $s(t)$ is a natural measure of cross-sectional price inequality: a wider spread between expensive and cheap properties gives a larger s and steeper slope.

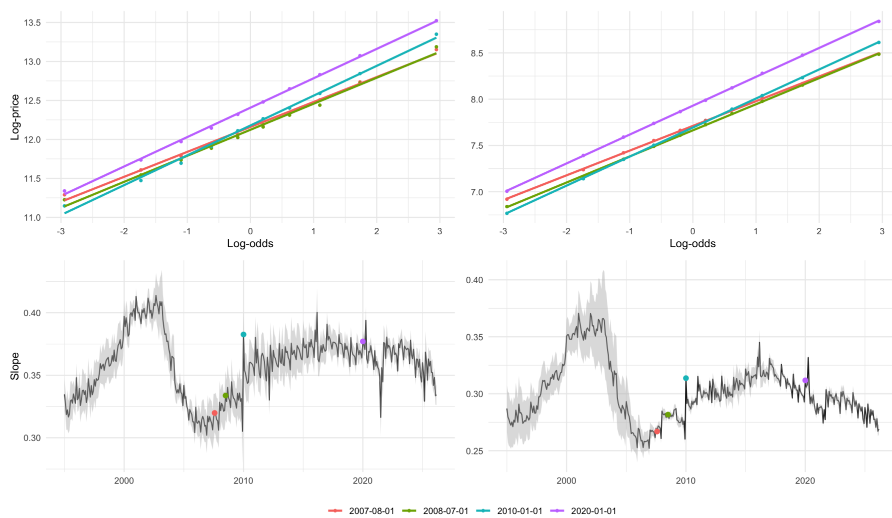

Figure 4: Top row: Decile log-odds versus log-price, at a selection of dates. Bottom row: Slope parameter with ± 2 x SE. By price (left column) and price per square metre (right column). Minimum R^2 values are 0.987 for the price dataset and 0.973 for the price per square metre dataset. The SE for the price per square metre dataset improves over time as the dataset itself improves (Figure A1).

The coordinate z is defined as

$$z = \log\left(\frac{q}{1-q}\right).$$

It carries a direct informational interpretation: as a log-odds, it is the scale on which statistical evidence, measured in nats, is additive, so equal increments of z represent equal increments of weight of evidence[17]. This is what makes the informational distance Δz used throughout (Section 3.7) an evidence measure.

3.3 Decomposition of the Growth Field

The growth field decomposes into exactly three parts. Differentiating the logistic structure gives

$$g(q,t) = \dot{\mu}(t) + \dot{s}(t)z(q,t) + s(t)\dot{z}(q,t),$$

with the ridge $r(q,t) \equiv s(t)\dot{z}(q,t)$. The first term is aggregate drift; the second, a reshaping (tilt) term, captures changes in the scale of the logistic profile; the third, the ridge, captures the motion of z. The drift is spatially uniform and the reshaping is linear in z, neither has an interior maximum.

The travelling wave is the ridge: a localised, single-peaked excess of relative growth, a departure from the instantaneous logistic fit, whose maximum defines the fastest-growing segment of the distribution; it is therefore a rank-space object

$$z^*(t) = \arg\max_z g(z,t).$$

Empirically it is a peak with progressively lower growth on either side (Figure 1) and is estimated as the per-month regression residual (Figure 5). We do not characterise its profile empirically; the simulation (Section 5) adopts a Gaussian, but the observed rank dynamics depend on the trajectory of the peak rather than on the profile's exact shape.

The log-price at this moving peak is $\ell^*(t) = \mu(t) + s(t)z^*(t)$. Differentiating along the trajectory $z^*(t)$ gives the same three contributions evaluated at the peak,

$$\frac{d\ell^*}{dt} = \dot{\mu}(t) + \dot{s}(t)z^*(t) + s(t)\dot{z}^*(t),$$

thereby separating the rate at which the entire distribution rises, the contribution of changing cross-sectional inequality, and the peak's intrinsic descent down the price ladder as it traverses the distribution. Under constant-speed advection (Section 3.4) the intrinsic descent term reduces to $s\dot{z}^* = -s\kappa$; Table 3 provides an empirical decomposition of the latter from the ensemble trajectories of Figure 2, Table 4 provides an empirical decomposition of all three terms from the same.

The reshaping term is itself a falsifiable prediction. Since the higher quantiles lead in the first half of a cycle and the lower quantiles in the second, the slope $s(t)$ should rise as the ridge approaches the median and decline thereafter, peaking at the median crossing. Figure 4 confirms this in both cycles, with peaks near 2002 and 2017, the same dates at which the wave peak trajectory crosses the median (Figure 2). We note that the slope increases from 2006 due to the wave superposition, as discussed in Section 2. The reshaping is consistent across cycles in aggregate: over the whole of each cycle, a longer time window than the active-traversal $\bar{s}$ of Table 3, the harmonic mean slope is 0.300 to the 2009 trough and 0.299 from it.

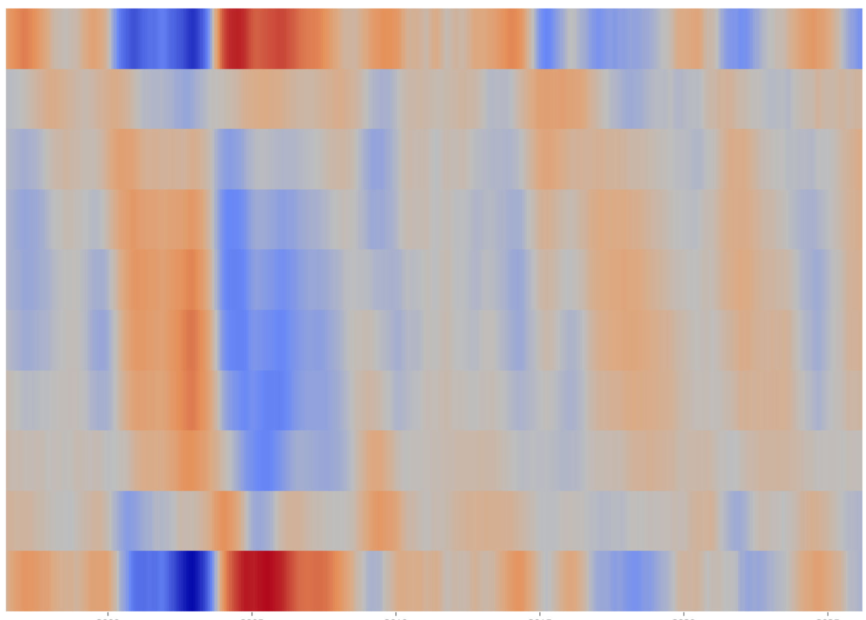

Figure 5: The growth ridge $r(q,t)$: the residual of the per-month regression of the 2×12-MA annual log return of the price per square metre dataset on the log-odds (Section 3.2). Highest decile at the top, lowest decile at the bottom. Above-trend growth in red, Below-trend growth in blue. The ridge is an instantaneous field, not a stock: the boundary accumulation of Section 4.2 concerns the capital density of Section 3.6, which is the time-integral of this field. Amplitude is cycle-dependent (Table 3). The first cycle's descent, its residence at the margin to 2008, the superposition at the top, and the second cycle's 'race for space', and the resumption at the lowest quantiles from 2023 are visible.

3.4 Uniform Advection in Standardised Log-Price Space

We model the ridge as a kinematic wave advected at constant speed in z:

$$\partial_t r - \kappa \partial_z r = 0, \qquad r(z,t) = R(z + \kappa t), \qquad z^*(t) = z_0 - \kappa t.$$

The ridge translates through the distribution without change of shape, and its peak $z^*(t)$ descends linearly at rate κ. The empirical linearity of these trajectories (Figure 2) is what allows us to treat κ as constant within a cycle; the window-sensitivity ranges reported there reflect the grid-search fitting rather than variation in κ; the simulation of Section 5 uses a single value for both cycles.

The adjacency evidence in Figure 6 shows that growth-rank transitions occur predominantly between adjacent price segments, which argues against a diffusive component and supports pure advection.

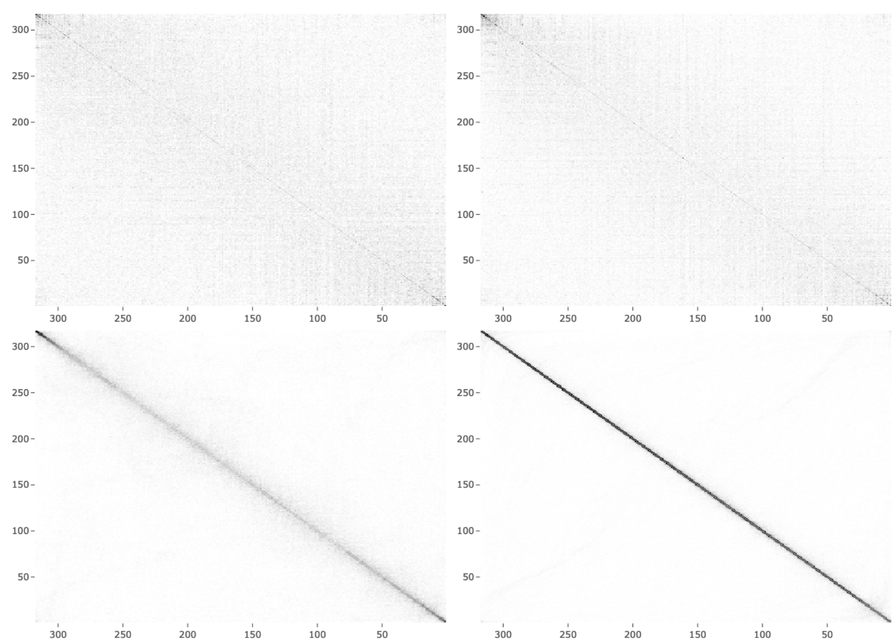

Figure 6: Adjacency matrices of the growth ranks; highest median price at the top, lowest median price at the bottom. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

The adjacency matrices are created from spatiotemporal networks of the growth ranks. Spatial connections link, at each date, segments holding adjacent growth ranks. Temporal connections link, for each growth rank, the segments holding that rank in successive periods. Both are combined into a single weighted adjacency matrix. Entry (i, j) records the normalised frequency with which segments i and j are linked by either connection type across the full sample. Since the matrices are constructed over identifiers ordered by price, diagonal concentration measures whether contemporaneous or successive relative growth leadership occurs among similarly priced identifiers.

There is a weak adjacency pattern for all Local Authorities (top row). However, there is a strong adjacency pattern for the quantiles (bottom row), even more so for the price per square metre dataset (bottom right). The strong diagonal concentration for the latter points to local propagation between adjacent quantiles. Furthermore, we conjecture that if it were possible to separate the land value from the overall house price (at any given quantile there will be a group of house prices with different mixes of land and materials), then the adjacency pattern would be stronger still.

Figure A22 discusses measures of diagonal concentration over time.

3.5 Mapping to Quantile Space

The mapping between standardised log-price and quantile space is nonlinear, with Jacobian

$$\frac{dq}{d\ell} = \frac{1}{s} q(1-q).$$

The ridge advection transforms as

$$\partial_t r - \kappa q(1-q)\partial_q r = 0,$$

with quantile-space velocity $-\kappa q(1-q)$, which integrates to the logistic trajectory

$$q^*(t) = \frac{1}{1 + \exp\left(\kappa(t - t_0)\right)},$$

the descending sigmoid of Section 2 (Figures 1 and 3). The wave begins at the highest quantiles ($q \to 1$) and ends at the margin ($q \to 0$).

The extended residence of the wave peak in the tails arises because $q(1-q)$ is small there, not because the propagation slows; the speed in z is undiminished. This is a geometric property of the logistic quantile map, not an imposed condition, and it is the same property that produces the accumulation of capital density at the margin (Section 4).

3.6 Competitive Reallocation of Capital

Let $f(q,t)$ be the capital density, predominantly mortgage credit, with $\int f dq = 1$. We choose two standard descriptions of competitive reallocation.

First, the replicator equation[18], from evolutionary game theory, has segments with above-average return gaining share at the expense of those below,

$$\partial_t f = f(g - \bar{g}),$$

which conserves total share by construction and requires no optimality assumption. With a prescribed growth field this has a standard closed-form solution $f \propto f_0 \exp(\int g dt)$ (normalised). Integrating and exponentiating the ridge field of Figure 5 gives the corresponding empirical quantity, the implied capital density of Figure A23. This exhibits a similar ordered cascade as that of the simulation of Section 5.

Second, a continuity equation,

$$\partial_t f + \partial_q J = 0, \quad \text{with Flux } J = vf,$$

also conserves total share, provided the flux vanishes at the boundaries. The adjacency evidence (Figures 6 and A22) favours the continuity description, in which capital moves locally, between neighbouring segments, rather than redistributing across the whole distribution.

These are different dynamics: the continuity equation does not follow from the replicator equation, both merely conserve total share.

We adopt the hypothesis that the capital density is advected by the same velocity field as the growth ridge, $-\kappa q(1-q)$. Under the log-odds transformation this continuity equation also retains a conservative form,

$$\partial_t f - \kappa \partial_z f = 0,$$

where f now denotes the capital density in z.

Economically, these dynamics are hypotheses about the cross-sectional structure of credit. The macro-financial literature[19] documents credit as endogenous to housing dynamics, but only in aggregate. Indeed, average mortgage credit per approval tracks median prices through both cycles (Figures A24 and A26), and the monthly change in credit leads the change in median prices outside the crash and recovery (Figure A27), consistent with mortgage lending as the vehicle of reallocation. However, direct evidence of a cross-sectional capital wave would require per-quantile lending data (Section 6.2).

Mortgages function as leveraged claims on land, and because land is a larger share of value at higher quantiles, leverage amplifies returns to land appreciation most strongly there; the land value gradient is thus a candidate explanation for why the ridge initiates at the top in both observed cycles.

A house price is not set by buyers alone but by the clearing of a leveraged transaction: across most of the distribution the purchase is credit-financed, so the price is where the supply and demand for mortgage credit meet. We suggest that the two conserved descriptions are these two sides. On the demand side, buyers direct credit-financed demand to the segment adjacent to the one they are priced out of: a local flux, the continuity description, and what the adjacency evidence (Figure 6) supports. On the supply side, lenders allocate credit across the distribution against portfolio- and market-level signals rather than street by street: the mean-field structure of the replicator equation, whose implied capital cascade (Figure A23) the data exhibit. Both sides are driven by the same travelling gradient in relative growth, which is why they share the velocity field κ. The cycle is the clearing path of credit supply and demand as the land value gradient is processed, and κ is the rate of that processing: the advective speed of the capital density, equivalently the speed of the ridge in standardised log-price space.

3.7 Cycle Time

The cycle lasts as long as the wave takes to traverse the price gradient: the geometry of the distribution fixes the informational distance Δz, and competitive reallocation fixes the rate κ at which it is processed,

$$T = \frac{z_{\max} - z_{\min}}{\kappa} = \frac{\Delta z}{\kappa}.$$

In quantile coordinates,

$$T = \frac{1}{\kappa} \int_{q_{\min}}^{q_{\max}} \frac{dq}{q(1-q)} = \frac{1}{\kappa} \log\left(\frac{q_{\max}}{1-q_{\max}} \cdot \frac{1-q_{\min}}{q_{\min}} \right).$$

The integral diverges as the limits approach the boundaries: the wave reaches $q = 0$ only asymptotically (Section 4.2), so the relevant Δz is set by where the distribution can be resolved, not by a literal arrival at zero land value. This is a data-resolution limit and not a free parameter, quantile estimates become unstable once a tail bin holds too few transactions. Figure A28 shows that $q \in [10^{-4}, 1 - 10^{-4}]$ corresponds to roughly 5–30 transactions per tail bin, the finest resolution at which monthly fits remain stable; 10^{-3} stops short of the resolvable margin and 10^{-5} outruns the transaction density, in both cycles.

With $\kappa = 1.1$, the three truncations give $T = 12.6$, 16.8, and 20.9 years. The data-justified choice (10^{-4}) gives 16.75 years; adding a crash-and-recovery interval brings this close to the $\sim$ 18-year periodicity reported by Foldvary and Harrison. Because T depends only logarithmically on the truncation, this is a consistency result. It is a range bracketing the observed cycle, with the central value fixed by transaction density rather than by fit to the prior literature, and not a precise prediction.

3.8 Logistic Factor

The factor $q(1-q)$ carries four coincident meanings, all specific to the logistic family: it is the Jacobian of the coordinate change; the velocity factor in the q-space dynamics; the logistic density in q-space; and the Fisher information of the below/above classification in the log-odds coordinate. This quadruple identity is what makes uniform advection in z and boundary-vanishing advection in q two views of the same dynamics, and why z is the canonical coordinate. The factor is formally identical to the $u(1-u)$ term in Fisher-KPP, reflecting the shared logistic structure, though its role differs: there it governs reaction in a frequency variable, here velocity in a quantile-space advection.

4 Crash Conditions

The framework derives the configuration in which a crash becomes possible, it does not derive the crash. We develop the condition in both coordinate systems of Section 3, which give complementary parts of the account. The z-space view fixes the timing: the wave consumes a finite informational distance at a constant rate. The q-space view fixes the mechanism and location: the boundary-vanishing velocity obstructs the conservative flux, so capital accumulates against the margin.

Converting that accumulation into a price correction requires one further premise, which the kinematics do not supply: that lenders and buyers respond to land values at the margin beyond what the margin can sustain. The configuration is endogenous; the trigger is assumed.

4.1 In z-space

The ridge peak is advected at constant speed κ through standardised log-price space (Section 3.4), and the capital density is transported with it (Section 3.6). The speed in z does not diminish as the boundary is approached, so the finite informational distance Δz to the resolvable margin (Section 3.7) is consumed in finite time, $T = \Delta z/\kappa$. Because the speed is constant, the time still to run is fixed by the wave peak's current position, $(z^* - z_{\min})/\kappa$; this is what makes the position a multi-year leading indicator (Section 6.3). It is the arrival of the configuration that is predicted, not the correction, which awaits the trigger.

Economically, this is the role Ricardo assigns to the margin: the binding constraint on the system. In his account the marginal land bears no rent and sets the level against which all other rents are measured; it is where the gradient organising the system is exhausted. Here the margin is the lowest-priced part of the distribution, and the wave's arrival there is the point at which the land value gradient that has organised the cycle has been consumed.

4.2 In q-space

Because the velocity is $-\kappa q(1-q)$ (Section 3.5), the flux $J = vf$ vanishes automatically at both boundaries, $J(0,t) = J(1,t) = 0$. This is a property of the log-logistic geometry, not an imposed condition. As the ridge peak approaches the margin, its leading-edge velocity vanishes while the trailing flux continues to deliver capital into the lower tail at the rate set by the wave's interior speed. By the continuity equation, capital density accumulates in a neighbourhood of the margin; Section A.4.8 provides a derivation. Under pure kinematics the accumulation is unbounded, because the idealised characteristics reach $q = 0$ only asymptotically. This asymptotic approach to the true boundary does not bear on the timing, the accumulation that constitutes the crash configuration builds during the approach and reaches crisis at the resolvable margin in the finite time T of Section 4.1.

Economically, the accumulating capital density is extended to the margin as the wave progresses, and the prices it supports carry a land value component beyond what the margin can sustain. The kinematics establish this accumulation but cannot by themselves reverse its sign. That reversal is the price correction, and it requires the premise above. Once lenders and buyers respond to the unsustainable values at the margin, capital and prices fall together, and the correction propagates back up the distribution.

Empirically, the correction did propagate up the land value gradient and attenuate with the rising share of land value at higher quantiles. Figure 1 shows that from spring 2008 the lowest-priced quantiles turn negative first, almost monotonically, the fall diminishing as it moves up the distribution. In Figure 4, the same is seen as the larger displacement at lower log-odds between 2007 and 2008. A repeat sales analysis (Figure A29) shows that the lowest decile and quantiles experienced the largest fall from their 2003–2007 peak to the 2009 trough. The boundary hypothesis is consistent with this sequence but, resting on one crash, is offered as a falsifiable prediction rather than an established result (Sections 6.2 and 6.3).

4.3 Framework Summary

Table 2 shows that the standardised log-price coordinate z and the quantile coordinate q are two views of the same dynamics.

Table 2: Summary of the framework by coordinate space.

Property	**Standardised log-price (z-space)**	**Quantile (q-space)**
Coordinate	$z = \log\left(\frac{q}{1-q}\right)$	$q \in (0, 1)$
Wave speed	κ	$\kappa q(1-q)$
Governing equations	$\partial_t r - \kappa \partial_z r = 0$ $\partial_t f - \kappa \partial_z f = 0$	$\partial_t r - \kappa q(1-q)\partial_q r = 0$ $\partial_t f - \kappa \partial_q [q(1-q)f] = 0$
Trajectory	$z^*(t) = z_0 - \kappa t$	$q^*(t) = \dfrac{1}{1 + \exp\left(\kappa(t - t_0)\right)}$
Cycle time	$\dfrac{\Delta z}{\kappa}$	$\dfrac{1}{\kappa}\displaystyle\int \frac{dq}{q(1-q)}$
Crash condition	Informational distance Δz consumed	Obstruction of conservative flux at the boundary

5 Simulation

In Figure 7, we verify that the kinematic model of Section 3, constrained to the empirically observed log-logistic distribution, reproduces the key features of the travelling wave documented in Section 2.

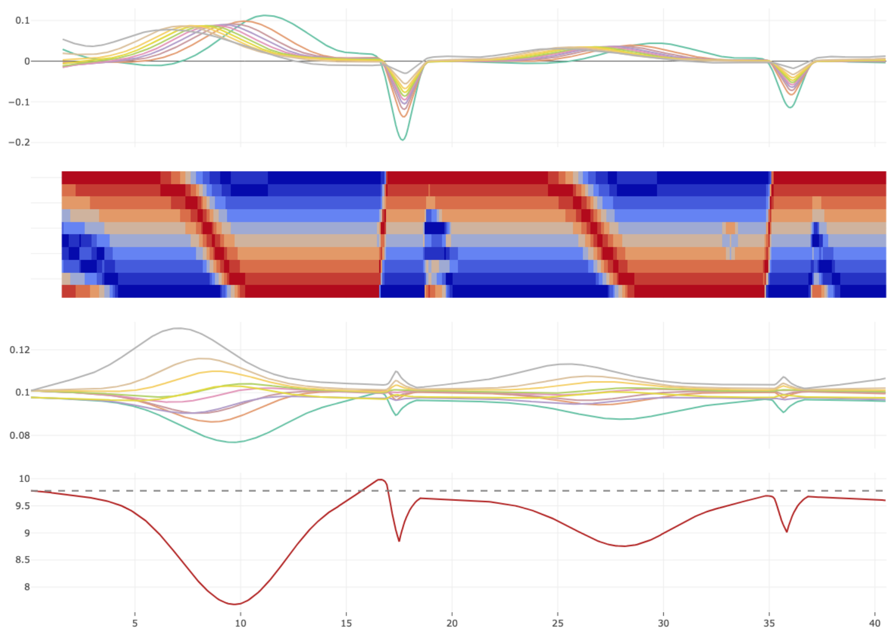

Figure 7: Simulated travelling wave. Top: Annual log return by decile. Heat map plot: Growth rank of the same by decile, with the highest median price at the top and the lowest median price at the bottom. Bottom: Capital density by decile (replicator dynamics, Section 3.6) and capital share of the bottom 10% of quantiles (dashed line: uniform-distribution share of the bottom band (31 of 317 quantiles, $\approx 9.8\%$)). The crash trigger is endogenous, as the wave reaches the boundary; its amplitude, duration, and recovery are imposed.

The simulation implements uniform advection of a Gaussian growth profile with a κ of 1.1 in standardised log-price space. Prices are initialised at the empirical cross-sectional medians from the first month of the price per square metre dataset, and the log-logistic price structure is maintained at each step by projecting onto the logistic manifold fitted to the current prices. Capital density evolves under the replicator dynamics of Section 3.6, with segments experiencing above-average growth gaining share at the expense of those below. The simulation uses 317 quantiles before aggregating to deciles.

The simulation reproduces three features of Section 1.4: monotonic propagation of the ridge, the logistic trajectory in q-space, and κ governing the propagation speed. The timing of the crash is endogenous, set by the wave's arrival at the distributional boundary; because we include no behavioural model, the occurrence of a correction at that moment is imposed, as are its amplitude, duration, and recovery.

The capital density plot exhibits the synchrony expected by construction (Section 3.6): the fractional share of each decile rises as the wave passes through it and falls thereafter, the replicator dynamics being driven by the same ridge. The peak of capital accumulation at the lowest decile coincides with the onset of the crash, the configuration in which conserved flux arrives at the distributional boundary. In the second cycle, with weaker growth, the accumulation does not exceed the uniform share: both simulated cycles use the same κ, so the wave reaches the margin on the same schedule, but the weaker cycle delivers less capital and so accumulates less by the time it arrives. The distinction between the wave's arrival at the margin and the build-up of sufficient density there is taken up in Section 6.3.

6 Discussion

In our framework, a Ricardian land value gradient (Section 1.3) under competitive reallocation (Section 3.6) gives rise to travelling waves.

6.1 Endogenous Origin

The travelling wave is a pattern in the ordering of relative growth rates, not in their magnitude. Three features of the evidence point to an endogenous origin. First, κ is approximately constant within each cycle despite substantial within-cycle variation in interest rates, credit conditions, and aggregate growth (Figure 2). Second, the wave appears independently within regions (Figure 3), making a single common national driver implausible. Third, the wave is robust to severe exogenous disruption: the pandemic temporarily displaced the ordering but it resumed at the same position afterwards (Figure 1).

Spatial correlation, differential macroeconomic sensitivity, and static responsiveness hierarchies can each generate persistent cross-segment growth differences, and simple directional coupling can produce monotonic propagation. What none reproduces is the logistic trajectory or the cycle time $T = \Delta z/\kappa$, against which we compare five classes of null model (Section A.5). The boundary accumulation of Section 4.2 follows from that trajectory, serving as a defining feature of the model rather than an outcome of a separate simulation.

Table 3: Empirical decomposition of the structural wave-speed quantities (price per square metre, deciles), across the 6,065 trajectories of Figure 2. Medians and interquartile ranges over the grid of windows; the IQRs are window-sensitivity ranges, not confidence intervals; ratios computed from unrounded values; terms in yr^{-1}. The empirical wave-peak velocity is decomposed over the same grid in Table 4.

Term	**C1 Med.**	**C1 IQR**	**C2 Med.**	**C2 IQR**	**Ratio C1/C2**
Drift, $\dot{\mu}$	0.140	[0.138, 0.141]	0.031	[0.028, 0.033]	4.53
Harmonic mean of the slope, $\bar{s}$	0.332	[0.326, 0.336]	0.312	[0.312, 0.313]	1.06
Traversal rate, κ	1.169	[1.077, 1.211]	1.030	[0.968, 1.056]	1.14
Intrinsic wave speed, $\bar{s}\kappa$	0.390	[0.351, 0.407]	0.321	[0.303, 0.330]	1.21

Aggregate growth determines how much capital flows through the housing market in a given cycle; competitive reallocation distributes this flow across the price distribution. The structural quantities, the traversal rate κ, and the slope $\bar{s}$ (both measured directly), and hence their product $\bar{s}\kappa$ respond far less to aggregate conditions than aggregate drift does. Table 3 shows that a $4.5\times$ difference in median drift between the cycles is associated with only $\sim 14\%$ variation in κ, $\sim 6\%$ in $\bar{s}$, and $\sim 21\%$ in $\bar{s}\kappa$.

Table 4 reports the peak log-price decomposition of Section 3.3. Each term, extracted from independently noisy monthly fits, emerges as a stable, correctly-signed quantity rather than as noise that cancels: the drift is positive, the intrinsic descent negative and dominant, and the reshaping term of smaller magnitude. Aggregate appreciation, inequality reshaping, and wave descent are each separately visible in the data.

Table 4: Empirical decomposition of the realised wave-peak velocity $d\ell^*/dt$ (price per square metre, deciles) into the drift, reshaping, and intrinsic-descent terms of Section 3.3, across the 6,065 trajectories of Figure 2. Medians and interquartile ranges over the grid of windows; the IQRs are window-sensitivity ranges, not confidence intervals; ratios computed from unrounded values; terms in yr^{-1}. The structural wave-speed quantities are decomposed over the same grid in Table 3 (construction in Section A.4.2), where $\dot{\mu}$ is the regression slope; the drift here is the window-mean rate, smaller by the curvature of μ.

Term	**C1 Med.**	**C1 IQR**	**C2 Med.**	**C2 IQR**	**Ratio C1/C2**
Drift, $\dot{\mu}$	0.127	[0.122, 0.133]	0.033	[0.030, 0.035]	3.85
Reshaping, $\dot{s}z^*$	0.050	[0.048, 0.052]	0.009	[0.009, 0.010]	5.36
Intrinsic descent, $s\dot{z}^*$	−0.338	[−0.352, − 0.324]	−0.301	[−0.313, − 0.285]	1.12

The terms differ markedly across cycles. The intrinsic descent is nearly unchanged (−0.338 and −0.301), whilst the drift has a difference of 3.85× (0.127 to 0.033). This realised intrinsic descent is the empirical counterpart of $-\bar{s}\kappa$ of Table 3 (−0.390, −0.321), the residual gap reflecting the constant-κ idealisation against the realised, time-varying $\dot{z}^*$.

The two observed cycles differ in κ in the same direction as their average growth, the weaker second cycle shows a lower traversal rate consistent with κ responding to the per-cycle conditions that set the gradient, even as it remains stable against the higher-frequency variation within each cycle. This relationship, and whether $\bar{s}$ is invariant across cycles, is a hypothesis for subsequent cycles, subject to constraints of Section 6.2.

The framework provides a common account of previously independent observations. The ripple effect (Section 1.1) can be interpreted as the geographic projection of quantile-space propagation when regions are ordered by price. The 18-year periodicity (Section 1.2) is consistent with $T = \Delta z/\kappa$; the cycle time is sensitive to both κ and the truncation of the quantile integral, with the 10^{-4} choice in Section 3.7 the finest truncation supporting stable estimation.

6.2 Limitations

The most fundamental constraint is sample size. The framework rests on just over thirty one years of transactional data across four representations and multiple spatial scales (Sections 2 and A.3). However, the travelling wave spans approximately 18 years per cycle, and the available data only partially cover two such cycles and one crash. The conclusions throughout are offered as falsifiable hypotheses for subsequent cycles rather than statistically established results.

The crash condition rests upon a single completed cycle. The first wave reached the lowest decile by approximately 2003 and was still propagating through the North West until 2008, with the correction following shortly after. Conventional accounts identify lending to borrowers at the margin as a proximate cause of housing crises, within this framework, that is what the wave reaching the boundary and capital accumulation looks like in practice. The boundary accumulation has not yet been tested by a second crash.

The economic mechanism proposed in Section 3.6, competitive reallocation of capital between adjacent price segments, is one explanation for the wave's propagation rather than a requirement of the framework: the predictions of Section 6.3 and the implications of Section 6.4 hold under any mechanism producing the observed wave behaviour. Per-quantile gross mortgage lending data, if made available by the Bank of England or UK Finance, would allow the cross-sectional distribution of capital flow to be derived; under the replicator dynamics, this flow should itself exhibit a travelling wave. Such data would test directly whether competitive reallocation operates at the segment level, closing the link to the capital density accumulation at the margin (Section 4.2) that the framework currently supplies only as a model-implied quantity. Per-quantile lending data may also be more readily available from many central banks than transactional data is from national land registries.

6.3 Predictions for This Cycle and the Next

The wave has resided at the lowest decile since the resumption following the pandemic. A simple 18-year cycle would place the crash in 2026, but the lower κ in the second cycle extends this, lower growth implies a slower accumulation at the boundary and a later date at which land values at the margin become unsustainable. Currently, the implied capital density does not show any accumulation (Figure A23). The previous correction was unambiguous in the data; we expect the next to be similarly so, initiating at the lowest-priced quantiles and propagating upward with attenuation. Should no capital accumulation and subsequent crash occur within the next couple of years, we would need to revisit the framework's assumptions.

We do not exclude the possibility of an exogenous shock forcing the behavioural response of Section 4, rather we note that for a crash to occur the system must be in the boundary accumulation state, with unsupportable land values at the margin.

The four contributions of Section 1.4 each generate a specific prediction for the next cycle. These are predictions about the wave's structural behaviour rather than the mechanism producing it. The wave peak will again propagate monotonically from highest to lowest decile (rather than randomly, or in reverse), with the ridge profile travelling alongside it (single-peaked and coherent, not random). The log-logistic price structure will continue to hold, with the slope $s(t)$ rising then falling as the wave traverses the median (Section 3.2). The within-cycle trajectory in z-space will again be linear, with κ constant (its window-sensitivity IQR again spanning only $\sim 10\%$ of the median, as in both observed cycles). And a correction will follow the wave peak's residence at the lower national margin, conditional on capital accumulation.

6.4 Implications for Land Value Taxation

We opened by noting that housing market crashes are often modelled as exogenous shocks. If the framework is correct, they are instead the terminal phase of an endogenous process, and the harm they cause is in principle addressable. This informs the long-standing debate over land value taxation[20].

Within the framework, a land value tax suppresses the cycle through two routes.

First, and more speculatively, it may lengthen the cycle. Cycle length is $T = \Delta z/\kappa$, and the informational distance Δz is fixed by the quantile geometry, independent of price levels, so cycle length responds to κ. The framework identifies κ as the rate at which competitive reallocation processes the land value gradient (Section 3.6). A land value tax falls by incidence on the land component of value, most where land's share is greatest, and so compresses the after-tax gradient; this weakens the differential that draws capital down the ladder, which would lower κ (Section 6.1) and lengthen T.

Second, and more robustly, it removes the boundary configuration al-together. The correction in Section 4 requires the wave to deliver capital to a margin whose land values have become unsustainable, and then a behavioural response to those values. A tax that prevents land at the margin from reaching unsustainable levels leaves that response with nothing to act on. This route does not depend on κ changing: even if cycle length were unaffected, the boundary accumulation that produces the correction would not arise.

The Georgist tradition has long held that land value taxation would suppress the housing cycle. Within this framework that claim is recovered as a consequence: through compressed dynamics if κ falls, and, more fundamentally, through the prevention of the boundary configuration in which the correction occurs.

Supplementary Material

A Additional Analysis

A.1 covers data and methods; A.2 exploratory data analysis; A.3 supporting analysis for Section 2; A.4 supporting analysis for Sections 3 and 4; and A.5 the null models.

A.1 Data and Empirical Methodology

A.1.1 Data Sources and Matching to Geographic Areas

England and Wales house prices from 1995 onwards are available from HM Land Registry Price Paid Data[21]. We also use a price per square metre dataset created by matching the PPD to EPC data, thereby controlling for property size. Our code[22] is a refactoring of the work by Chi et al.[23]. We suggest that the price per square metre dataset is the best available proxy for land value.

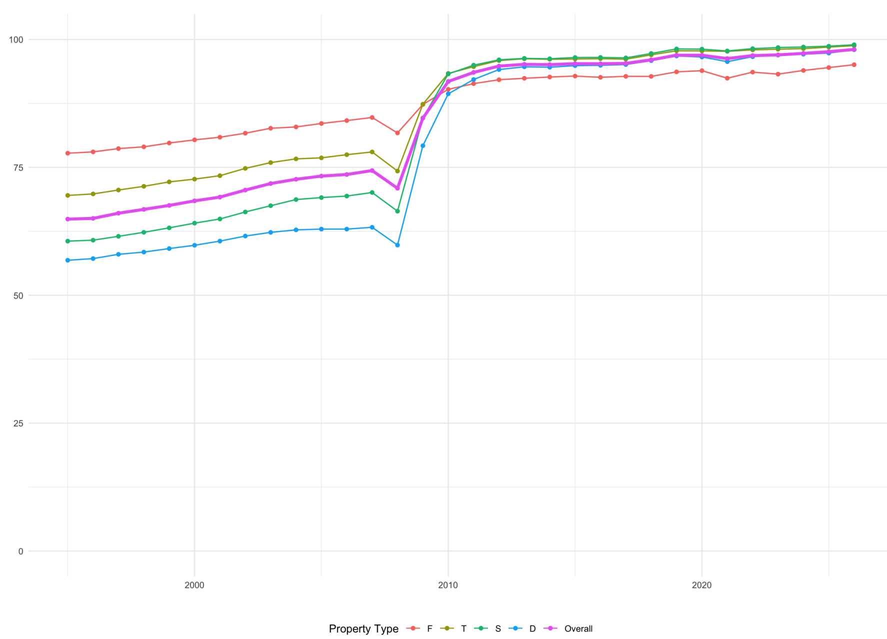

Figure A1: Price per square metre dataset match percentage per year by property type, and overall.

From the Price Paid Data and price per square metre dataset, we exclude property type transactions representing the 'Other' category, leaving records for detached, semi-detached, terraced, and flats/maisonettes. We also exclude the Additional Price Paid category ($\sim 3\%$ of transactions), as we do not know the additional price.

We match to a Local Authority code using the ONS Postcode Directory[24], and then to a Local Authority District[25] and Region[26]. Additionally, we have created a manual mapping of County and District to Local Authority and Region for those records without a Postcode or where there is no match in the ONS PD. The Isles of Scilly are excluded as there are too few sales. We exclude data beyond March 2026 due to a lag in registrations at Land Registry.

A.1.2 Stamp Duty

A cut in Stamp Duty is often assumed in the popular press to leave house prices unchanged; it does not. As RICS guidance[27] explains, "land is the residual" since the supply of land is inelastic. Hence, the proper price of a property is the price including Stamp Duty. As Adam Smith noted, "He considers what the land will cost him in tax and price together. The more he is obliged to pay in the way of tax, the less he will be disposed to give in the way of price."[28]

We are not able to determine whether a transaction is for a second home at a higher rate of Stamp Duty, so we apply the basic rates to all. Stamp Duty has been calculated using historical rates, and based on the highest band the price falls into (a discrete tax regime, before December 3, 2014), or based on the portion within a given band (a marginal tax regime, thereafter). Similarly, values for England and Wales, England alone, or Wales alone have been used, as appropriate.

We do not consider the impact of the High Value Council Tax Surcharge.[29]

A.1.3 Measurement of Central Tendency and Ordinal Ranking of Growth

We analyse the change in median house prices over time. Given the wide range of house prices and compositional change each month per Local Authority, the median is chosen as the best measure of central tendency. The annual log return of the median is calculated per Local Authority, and a 2×12 moving average (13-month symmetric) adjusts for seasonality. The code also includes the option to use an 11-month (plus the current month) trailing moving average for prediction. Partial computations are allowed, as such the moving average is less robust at the beginning and end of the time series. For Ricardo's cascade of the change in rent, we calculate the ordinal ranking of the growth rates.

An analysis by empirical quantiles (equal-mass rank bins) is then conducted for both datasets, with the number of quantiles matching the number of Local Authorities.

Because quantiles are recomputed each month, the price at a given quantile tracks an order statistic of a changing transaction mix rather than a fixed set of properties, and shifts in the composition of sales can in principle move quantile prices with no like-for-like change in values. Three features of the analysis limit its influence. First, the travelling wave appears (more noisily) in the Local Authority representation (Figure A13), whose units are fixed by geography and cannot migrate across the ordering. Second, the price per square metre dataset controls the largest within-unit compositional dimension, floor area. Third, the pattern appears independently within each property type (Figure A19), excluding type-mix shifts as a driver. The quantile representation is therefore a change of coordinates whose results are corroborated by representations immune to compositional reshuffling.

A.1.4 Alternative Time Series Decompositions

Seven different time series decomposition approaches were investigated. Although the respective trends of median price obtained were always smooth, this was at the cost of placing too much signal into the random / remainder / irregular / residual categories, especially during the 2007-2009 crash. Code is available for SEATS and LOESS plots which show the same pattern as the simple moving average approach.

A.1.5 Heat Map Visualisations

Heat maps are created of the median price per Local Authority or quantile over time; we order the Local Authorities from high to low using the median price from the last year of data. Local Authorities are somewhat spatially autocorrelated, so such an ordering is the simplest possible approach. The Local Authority plots are based on a similar heat map (Figure B1) from 2017 by Neal Hudson of consultancy Built Place. Whereas Hudson's heat map also shows an overlay of the single fastest-growing Local Authority over time, we create a separate plot showing all growth ranks for Local Authorities or quantiles. We then group by deciles to reduce the noise in the plots.

A.1.6 Credit

The only mortgage data available is for 'Value of residential mortgage loans outstanding in the UK, split by sector postcode'[30]. This data includes house purchases, equity release, remortgaging, and repayments. It is not a like-for-like comparison with sales transactions.

Instead, the following pairs of data from the Bank of England[31] have been used:

- LPMB3C2 Monthly values of total sterling approvals for house purchase to individuals (in sterling millions) not seasonally adjusted
- LPMVTVU Monthly number of total sterling approvals for house purchase to individuals not seasonally adjusted
- LPMB3XF Monthly value of monetary financial institutions' sterling gross approvals for house purchase to individuals (in sterling millions) not seasonally adjusted
- LPMB3ZF Monthly number of monetary financial institutions' sterling gross approvals for house purchase to individuals not seasonally adjusted
- LPMBV86 Monthly value of total sterling gross approvals for house purchase to individuals (in sterling millions) not seasonally adjusted
- LPMBV87 Monthly number of total sterling gross approvals for house purchase to individuals not seasonally adjusted

Added to each value series is the Help to Buy data[32].

A.2 Exploratory Data Analysis

We undertook analysis of the probability density functions and log returns to get a feel for the data and the direction of subsequent analysis.

A.2.1 Probability Density Functions

Figure A2 shows the extent to which house prices are rounded to the nearest large number, with the highest peaks highlighting the distortion of the old Stamp Duty Regime.

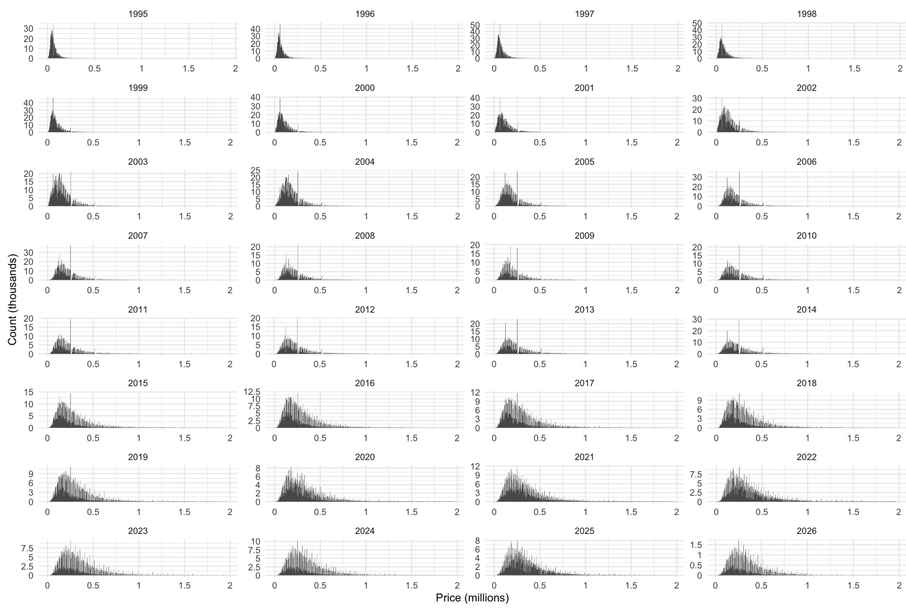

Figure A2: Price pdf (histogram), by year.

Figure A3 shows that the price per square metre distribution is far more continuous.

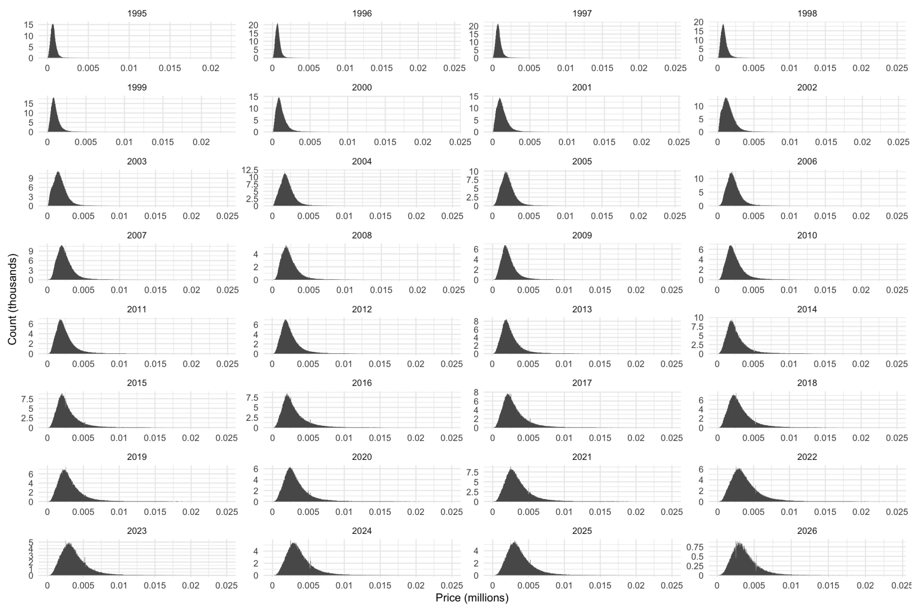

Figure A3: Price per square metre pdf (histogram), by year.

In Figure A4, the Cullen-Frey plot shows that the probability density functions are log-logistic, since the log-price is logistic with only minor skew.

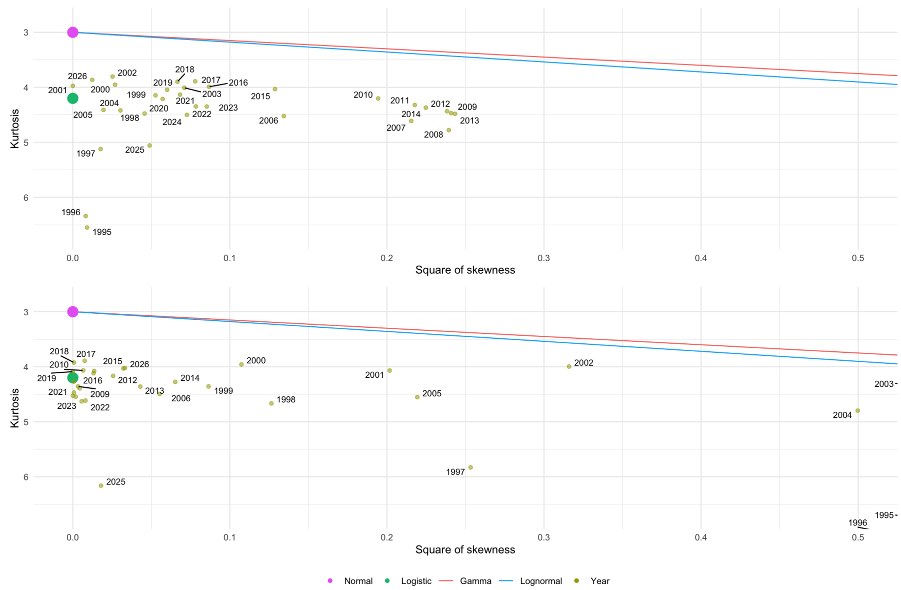

Figure A4: Log-price kurtosis and square of skewness, by year. Top: by price. Bottom: by price per square metre.

A.2.2 Local Authority Probability Density Functions

We investigate the possible candidates for the house price distribution by Local Authority.

Figure A5 shows the wide variety of price distributions per Local Authority, justifying the use of the median as a measure of central tendency. The price per square metre dataset has far less kurtosis and skew.

The log return plots show concentration around normal and logistic distributions. We analyse the log returns in Section A.2.3.

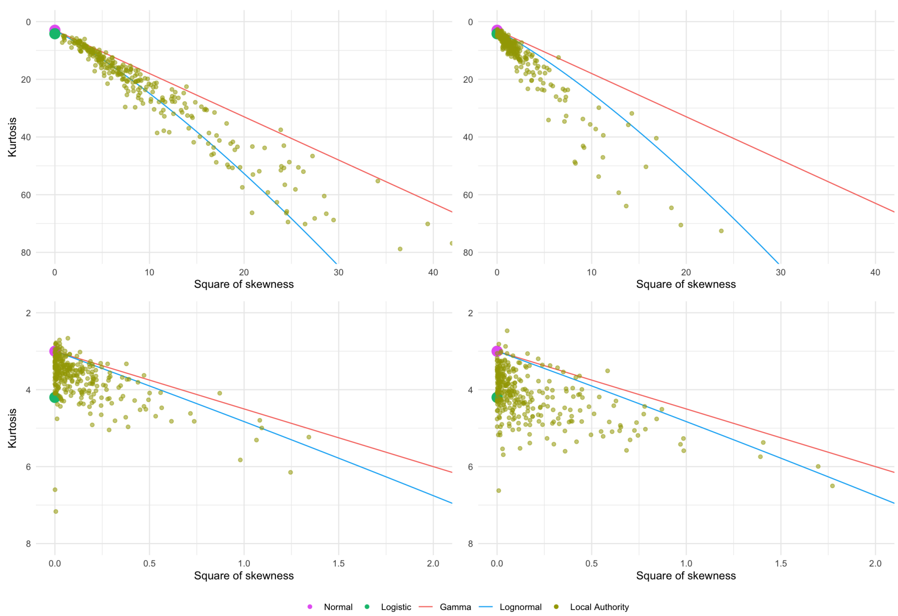

Figure A5: Top: Price pdf kurtosis and square of skewness, by Local Authority (2025 onwards). Bottom: Annual log return pdf kurtosis and square of skewness, by Local Authority (all years).

A.2.3 Log Return Analysis

In Figure A6 we plot the log returns over the entire time period. Local Authority log returns aggregate growth across many somewhat independent regional distributions and are therefore approximately symmetric. In contrast, quantile log returns describe the movement of ranks within a single evolving distribution and therefore do not exhibit symmetry.

The spikes at zero log return for the price distribution (left column) suggest that markets regularly stay flat month-on-month.

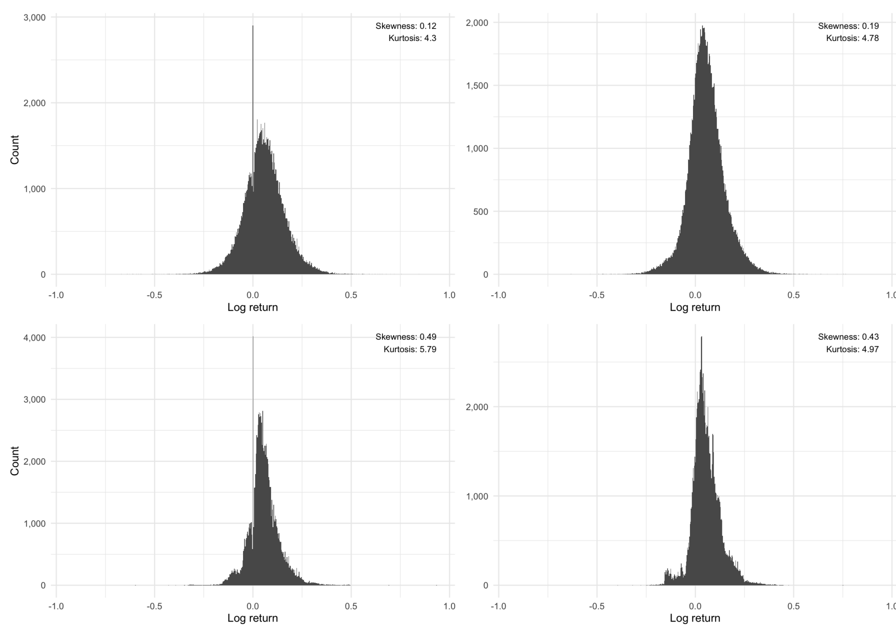

Figure A6: Log return of the median price. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

In Figure A7, the mean and standard deviation of the log returns are plotted for the entire time series. The triangle-shaped kinks in the quantile price plot are due to the old discrete Stamp Duty regime.

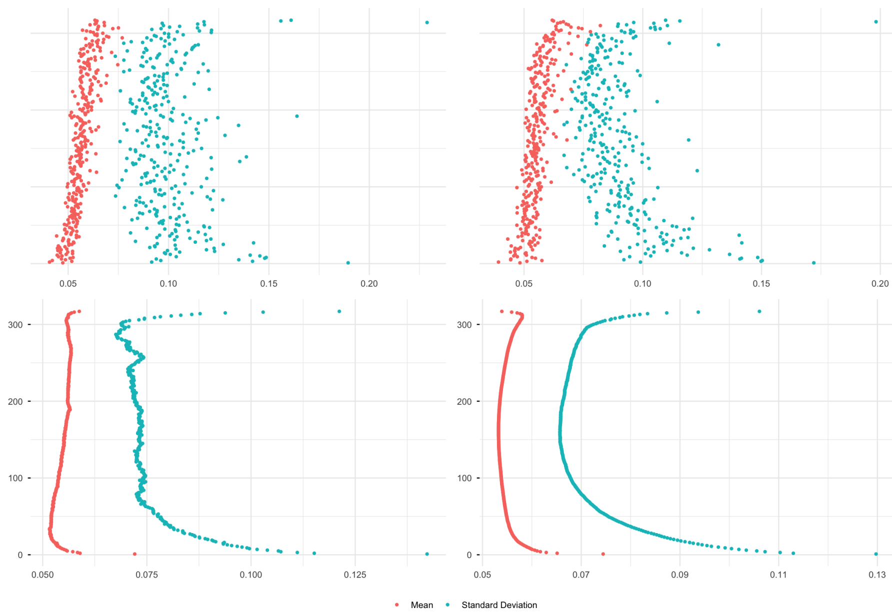

Figure A7: Mean and standard deviation of the log return of the median price; highest median price at the top, lowest median price at the bottom. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

The standard deviation of the tails in the quantiles plots is significant and leads to four clusters in the PCA K-means clustering plot in Figure A8. From this analysis, the north/south housing divide (Figure A9) is derived.

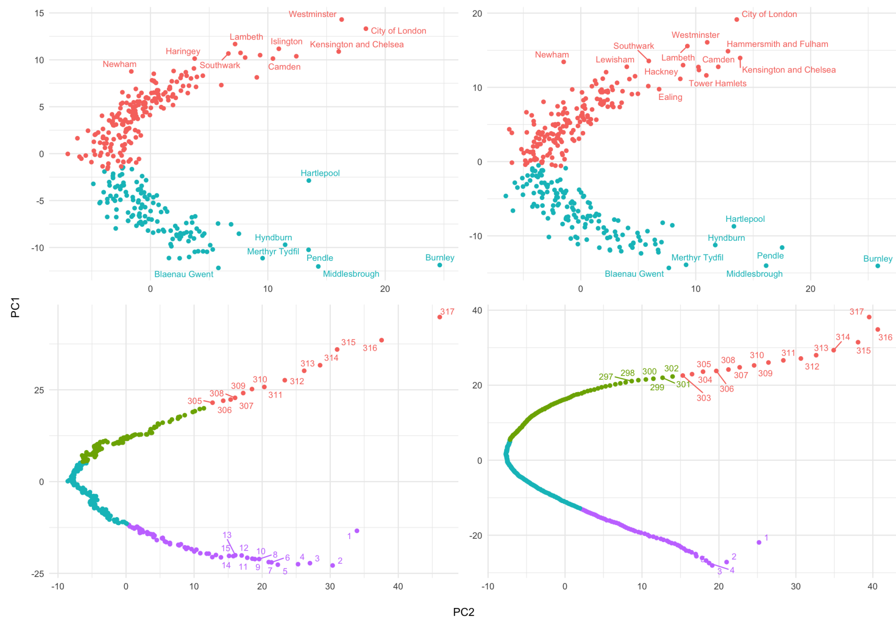

Figure A8: Principal Component Analysis K-Means Clustering. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right). PCA axes have been flipped to align with the convention of Local Authorities and quantiles on the y-axis, with the highest median at the top.

In Figure A9, using the ONS shape file[33], a map of the north/south divide, which is not that different from the work of Danny Dorling[34].

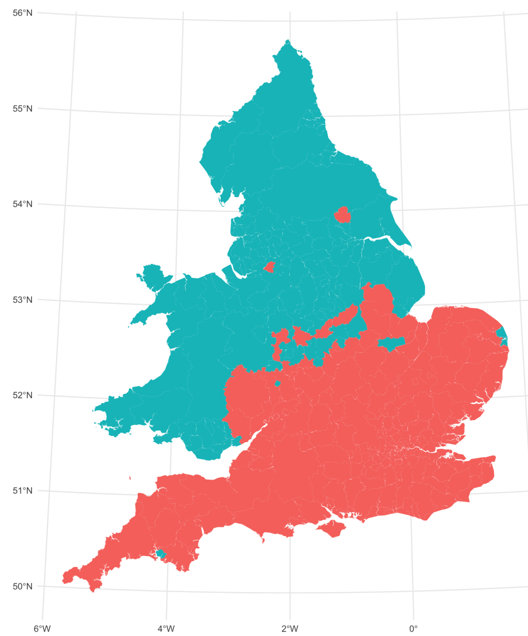

Figure A9: North South divide, K-Means clustering (n = 2) of log returns.

For comparison, in Figure A10, a map of the log median price.

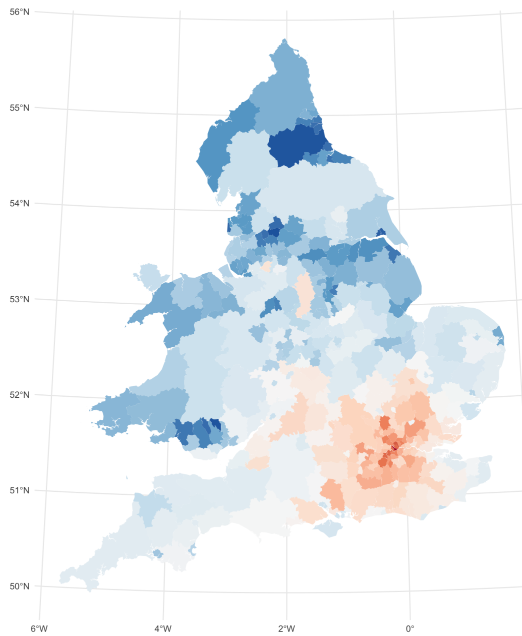

Figure A10: North South divide – log median price.

And using three clusters, Figure A11 separates out central London, plus Richmond and Elmbridge.

Figure A11: Principal Component Analysis, K-Means Clustering (n = 3), by Local Authority.

A.3 Empirical Results and Robustness Testing

A.3.1 Annual Returns

Figure A12 shows the moving average of the annual log return of house prices. The top row is by Local Authority, the bottom row by quantile. The left column is calculated from price, the right column from price per square metre.

In the cycle to 2007, median house prices initially increased in the highest-priced areas, then over time increased across the middle and lower-priced areas. The crash is very clear in blue across all Local Authorities and quantiles.

In the bottom left quantile plot, we see stepped patterns due to the old discrete Stamp Duty regime, these correspond to the thresholds of £60,000, £250,000, and £500,000.

From 2010, there is the start of a recovery before a modest fall in prices in the austerity years. The pattern starts again afterwards but without such high log returns as in the first cycle. Later there is the impact of the pandemic and the subsequent increases in property prices before the aftermath of the 2022 mini-budget.

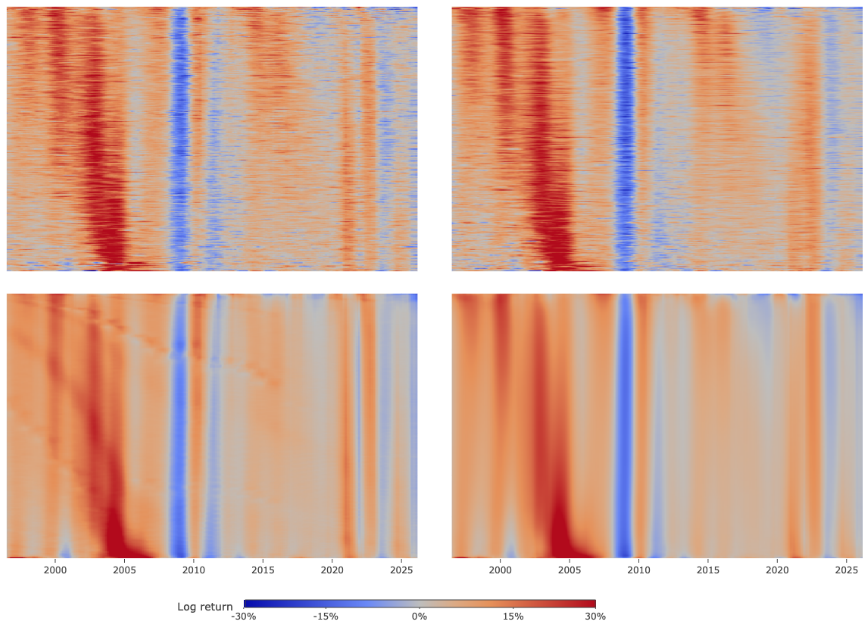

Figure A12: Moving average of the annual return of the monthly median; highest median price at the top, lowest median price at the bottom. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right). Colour scale fixed to ±30% to increase the contrast.

A.3.2 Growth Rank of the Annual Returns

Figure A13 shows the growth rank of the log returns over time, allowing us to visualise the change in house prices independently of the underlying house price growth. Colour scale as in Figure 1.

In all plots, there are clear slanted red lines representing the fastest-growing areas in the house price cycle to 2007 and in the cycle from 2010. (In the bottom left quantile plot, we once again see stepped patterns due to the old Stamp Duty regime.) The rate of progress is very similar in both cases.

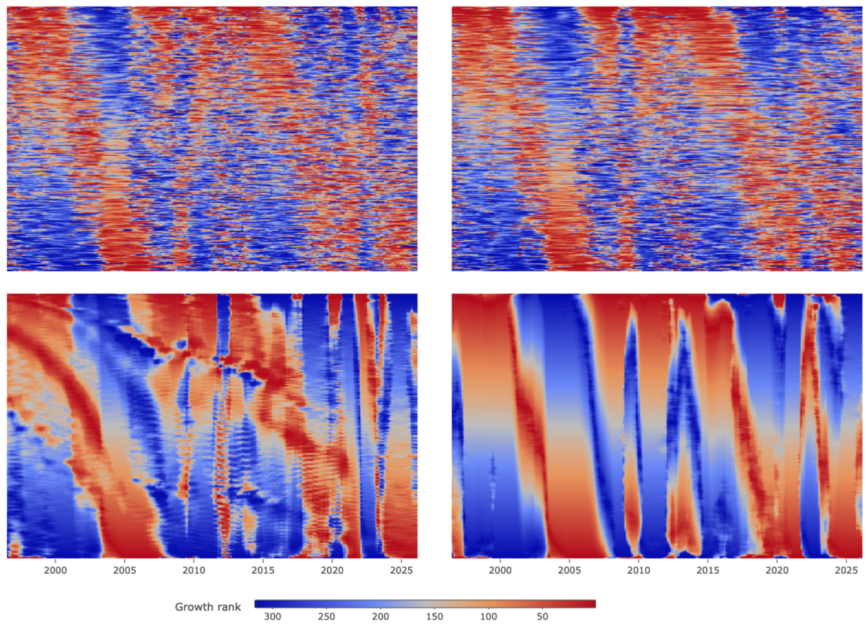

Figure A13: Growth rank of the moving average of the annual log return of the monthly median; highest median price at the top, lowest median price at the bottom. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

Notably, the travelling wave reaches the Local Authority or quantile with the lowest median price prior to 2007. However, some of the lowest-priced Local Authorities, such as Burnley, Pendle, and Blackburn with Darwen, were among the fastest growing nationally, confirming that the wave peak had propagated through the full distribution. The apparent new wave visible in the richest areas immediately prior to the crash is a superposition (Section 2.2), not a restart; the mature wave subsequently reached the margin.

In Figure A14, we present the growth rank plot but allowing the ordering of the Local Authorities to be re-ordered every month. This allows for areas that gentrify to move up the distribution.

The previous horizontal lines are lost in favour of a noisier pattern. Although the number of pixels has not changed, the impression is that the resolution has increased.

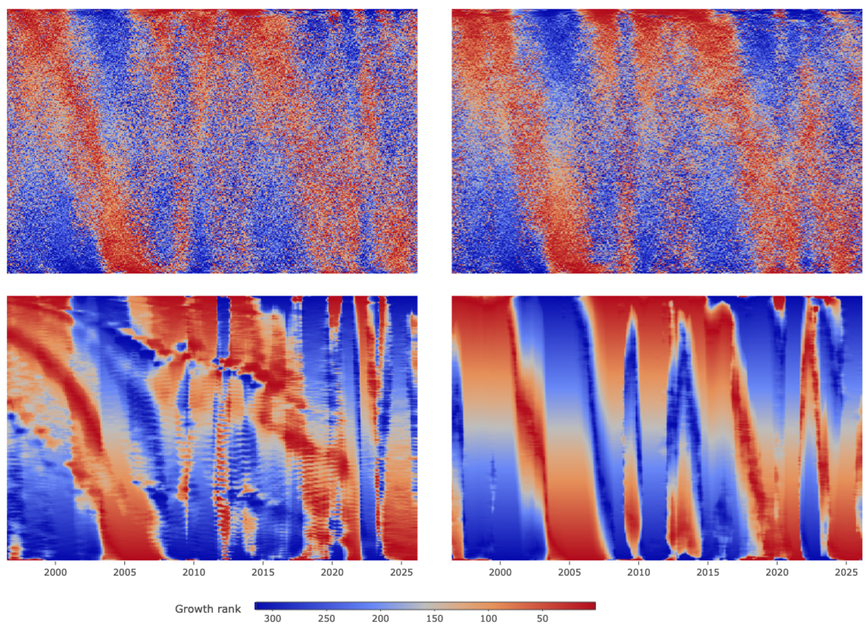

Figure A14: Growth rank of the moving average of the annual log return of the monthly median; highest median price at the top, lowest median price at the bottom. By monthly re-ordered Local Authority (top left), by monthly re-ordered Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

In Figure A15, we analyse how Local Authorities change their relative rankings over time. We plot Spearman's rank coefficient, along with a comparison of the initial ranking versus final ranking.

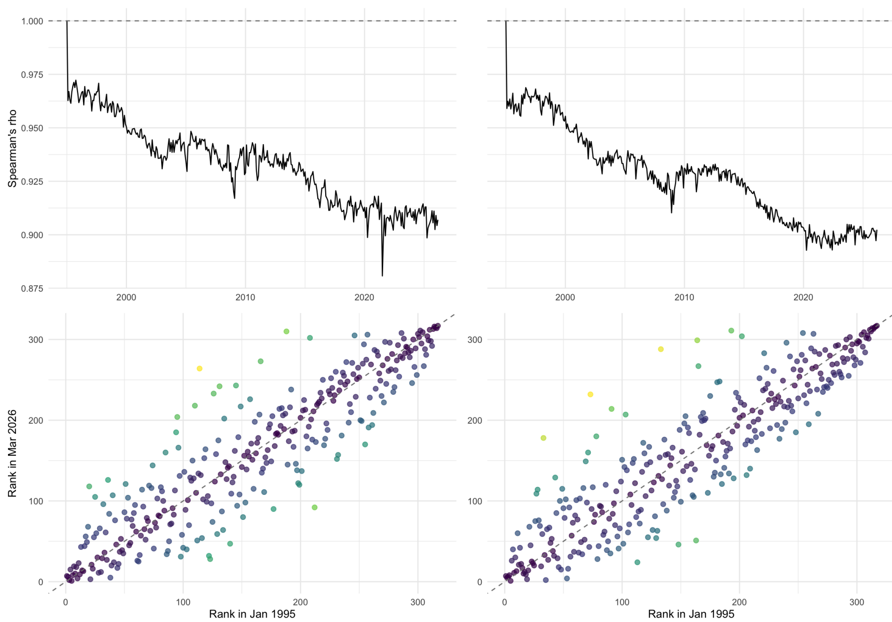

Figure A15: Top row: Local Authority Spearman rank coefficient. Bottom row: Rank comparison of first and final months. By price (left column) and price per square metre (right column).

Figure A16 repeats Figure A13 but using deciles.

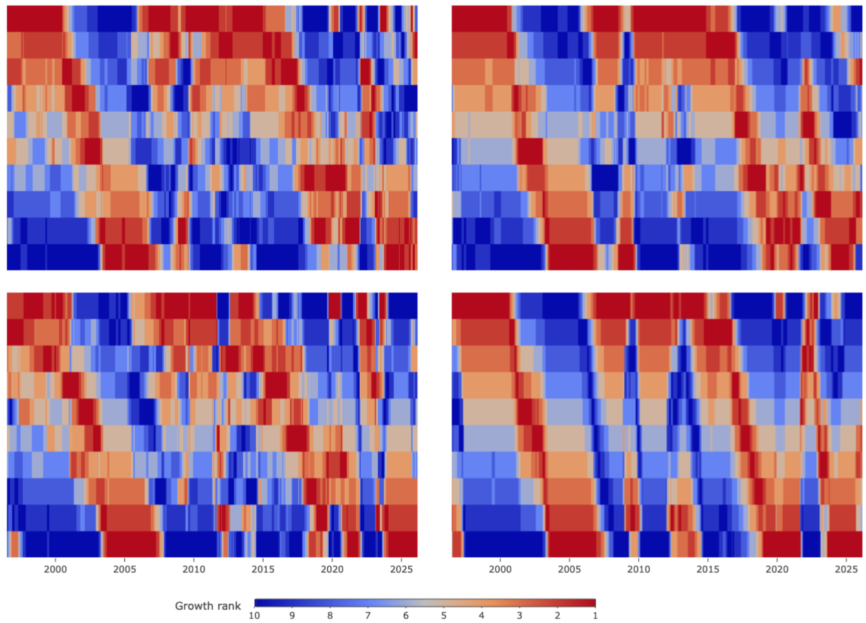

Figure A16: Growth rank of the moving average of the annual log return of the monthly median, by decile; highest median price at the top, lowest median price at the bottom. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

In Figure A17, we present the un-smoothed decile growth rank plots. The overall patterns are still visible.

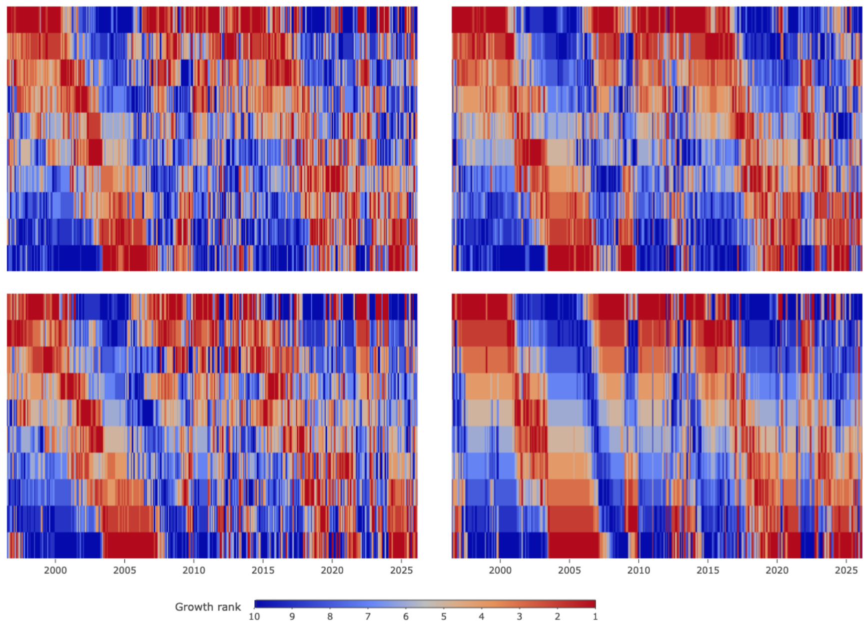

Figure A17: Growth rank of the annual log return of the monthly median, by decile; highest median price at the top, lowest median price at the bottom. By Local Authority (top left), by Local Authority using the price per square metre dataset (top right), by quantile (bottom left), and by quantile using the price per square metre dataset (bottom right).

A.3.3 By Region and Property Type

Figure A18 shows a regional decomposition of the growth rank plot. We note the cascading travelling waves over time. Unfortunately, the second cycle is not as clear.

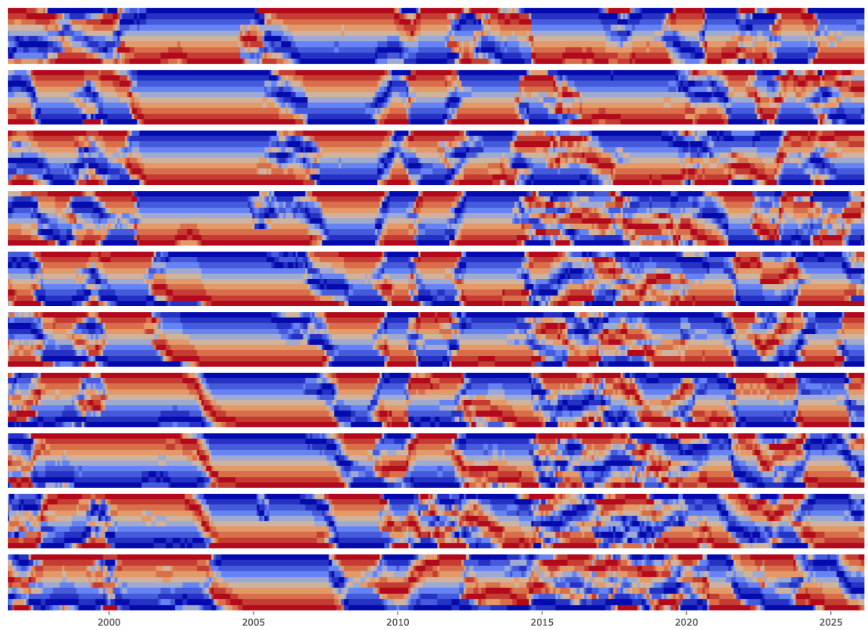

Figure A18: Growth rank of the annual log return of the monthly median, by decile; regions arranged by descending median price: London, South East, East of England, South West, West Midlands, East Midlands, Yorkshire and The Humber, North West, Wales, North East.

In Figure A19, we show a decomposition by property type of the growth rank plot. We once again see superposition of a new travelling wave prior to the crash. There does not seem to be any bias due to lower match rates (Figure A1) prior to 2010.

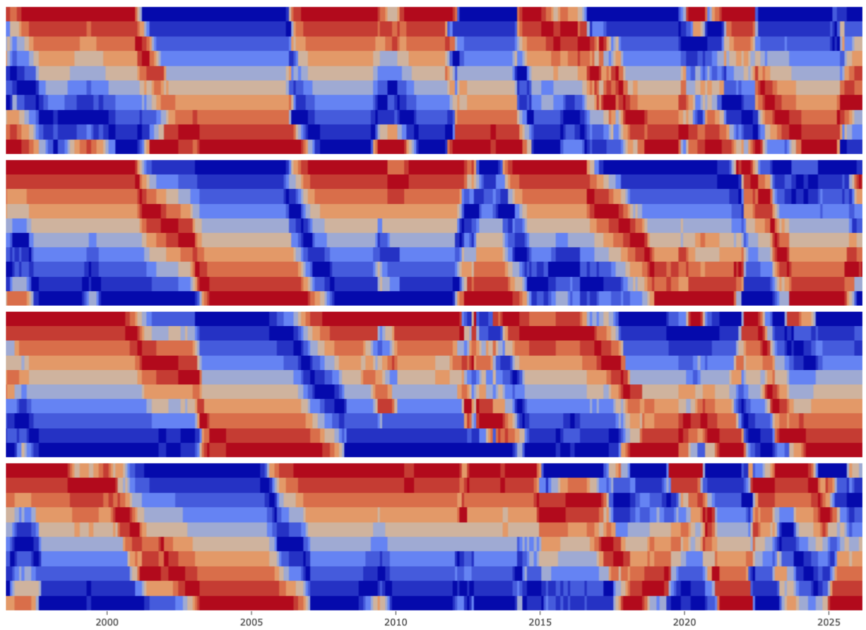

Figure A19: Growth rank of the annual log return of the monthly median, by property type: detached, semi-detached, terraced, and flats/maisonettes.

A.3.4 Growth Rank in z-space

In Figure A20, we show the z-space representation.

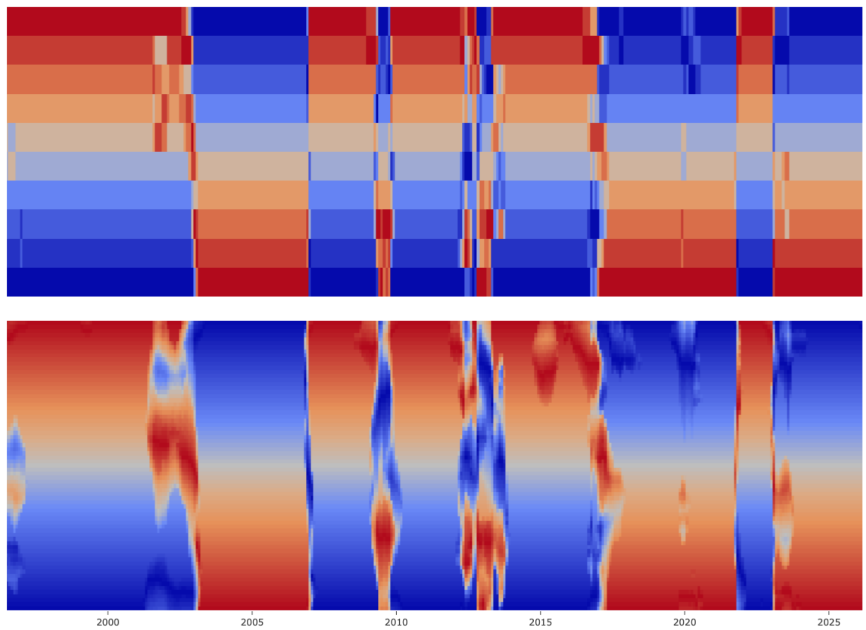

Figure A20: Growth rank in standardised log-price space. Top: Equal z-interval bins (width 0.7). Bottom: Nadaraya-Watson kernel-smoothed continuous surface (Epanechnikov kernel, bandwidth 1).

The z-space representation confirms the travelling wave is robust to binning scheme. The temporary disruption from the pandemic 'race for space' and the subsequent resumption of the wave peak at the same standardised price level is particularly clear in this representation.

The extended residences at the distributional extremes are a feature of the logit coordinate stretching sparse q-regions, not a separate dynamical regime.

The kernel-smoothed surface is robust to kernel choice: Gaussian, Epanechnikov, and triangular kernels at equal bandwidth produce very similar results.

A.4 Travelling Wave Model – Supporting Analysis

A.4.1 Logistic Log-Price Structure

In Figure A21, we provide the log-price versus log-odds for the full 317 quantiles. Minimum R-squared values are 0.982 for the price dataset and 0.972 for the price per square metre dataset.

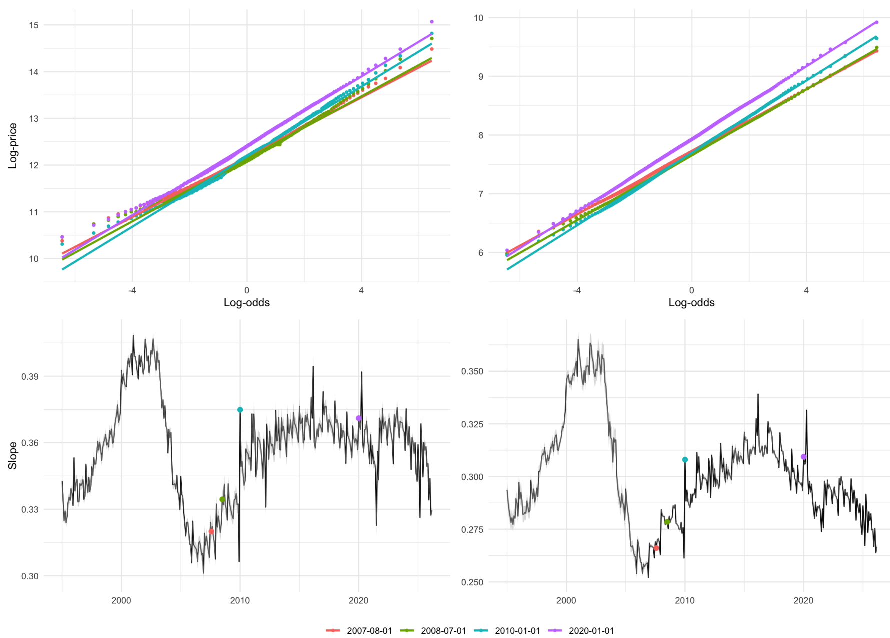

Figure A21: Top row: Quantile log-odds versus log-price, at a selection of dates. Bottom row: Slope parameter with $\pm 2\times$ standard error band. By price (left column) and price per square metre (right column).

A.4.2 Decomposition of the Wave Peak Velocity

The decomposition of Table 4 is formed point-wise from the monthly series and aggregated over the same grid of windows as Table 3. The peak position z^* is taken from the 2×12-MA growth field; the slope $s(t)$ is also calculated as a 2×12-MA series before differencing, so that both factors entering $\dot{s}z^*$ and $s\dot{z}^*$ share the basis on which z^* is defined. On the raw, un-smoothed slope the reshaping term $\dot{s}z^*$ is dominated by month-to-month fitting jitter while the drift and intrinsic-descent terms are essentially unchanged; the smoothed slope is therefore used throughout.

Each term is the cycle mean of its monthly series; summed within each window, the three reproduce that window's measured velocity $d\ell^*/dt$ to within 0.0015 yr^{-1} across the grid (smoothed variant; the medians need not sum to the median velocity), the remainder a second-order differencing term. The drift on this basis is a mean of month-to-month changes, that is, the net change in μ over the window divided by its length, and has a value below the regression-slope $\dot{\mu}$ reported in Table 3, the basis on which κ, and hence $\bar{s}\kappa$, is fitted. The two coincide only when the trajectory is straight and differ here by the curvature of $\mu(t)$ over the window.

A.4.3 Adjacency and Local Propagation

Figure A22 shows the diagonal concentration over time for 5% and 10% bandwidth.

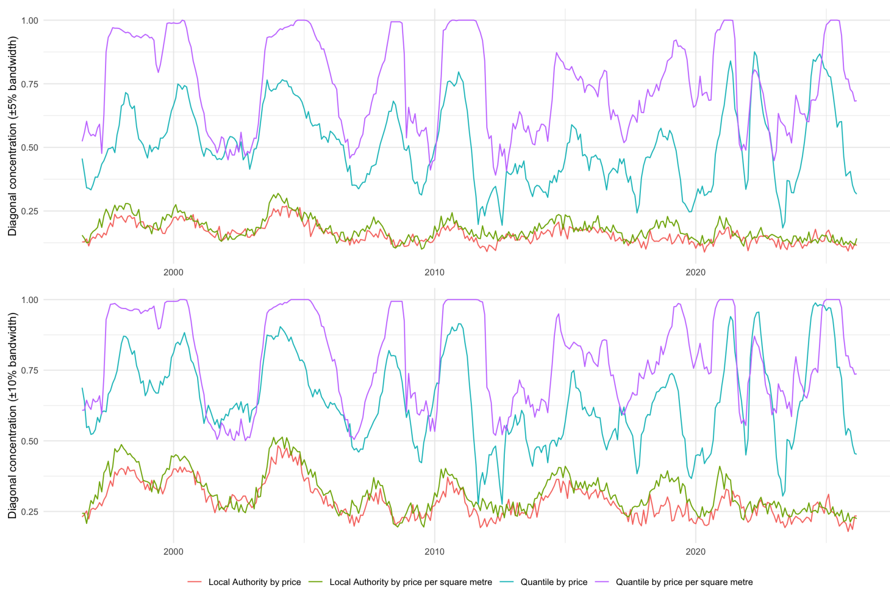

Figure A22: Diagonal concentrations over time, at 5% and 10% bandwidth. By Local Authority, by Local Authority using the price per square metre dataset, by quantile, and by quantile using the price per square metre dataset.

A diffusion-dominated evolution would be expected to produce gradual erosion of price-local growth adjacency. Empirically, diagonal concentration remains elevated within regimes and declines primarily at points where relative growth leadership shifts across the price ordering, after which locality is rapidly re-established. This pattern is consistent with discrete reassignment of relative growth leadership across price tiers rather than gradual dispersion, and does not require the introduction of a diffusive component to the advection model.

Because adjacency is measured in rank but locality is a price property, the rank-to-price map's boundary behaviour governs whether the two coincide across the distribution. The preservation of diagonal concentration under bandwidth relaxation suggests that they do, consistent with quantile maps exhibiting symmetric first-order boundary divergence (Section A.4.7).

A.4.4 Replicator Dynamics and Implied Capital Density

Figure A23 shows the model implied capital density from Section 3.6.

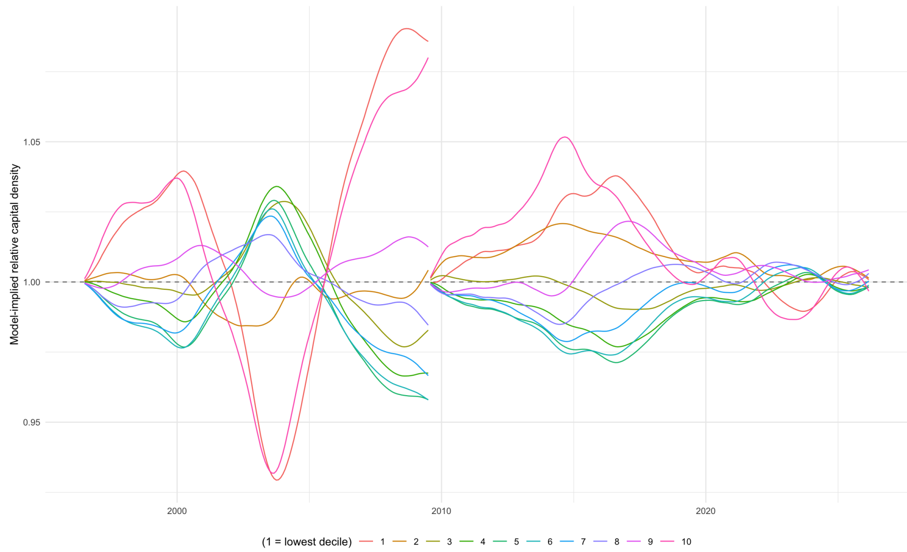

Figure A23: Model-implied relative capital density by decile, the empirical counterpart of the capital density panel of Figure 7. Under the replicator reading the capital density is proportional to $\exp(\int r dt)$. Each decile relative to the cross-sectional mean (dashed line). The integral is restarted in late 2009, since it is model-meaningful only within a cycle; the 2×12 smoothing makes it approximate. This is a model-implied quantity, not a measured capital, which awaits the per-quantile lending data of Section 6.2. Both cycles reproduce the ordered cascade of the simulation, they also show the superposition and the 'race for space' interruption in each cycle respectively. There is clearly accumulation at the margin in the first cycle but not (yet) in the second cycle.

A.4.5 Credit versus Median Prices

Figure A24 shows median house prices align with the amount of credit per approval until the pandemic. At different times, there will be variance in the equity mix and number of cash buyers. Mortgage credit is itself created when loans are issued and destroyed when they are repaid[35]. Also shown is cumulative credit.

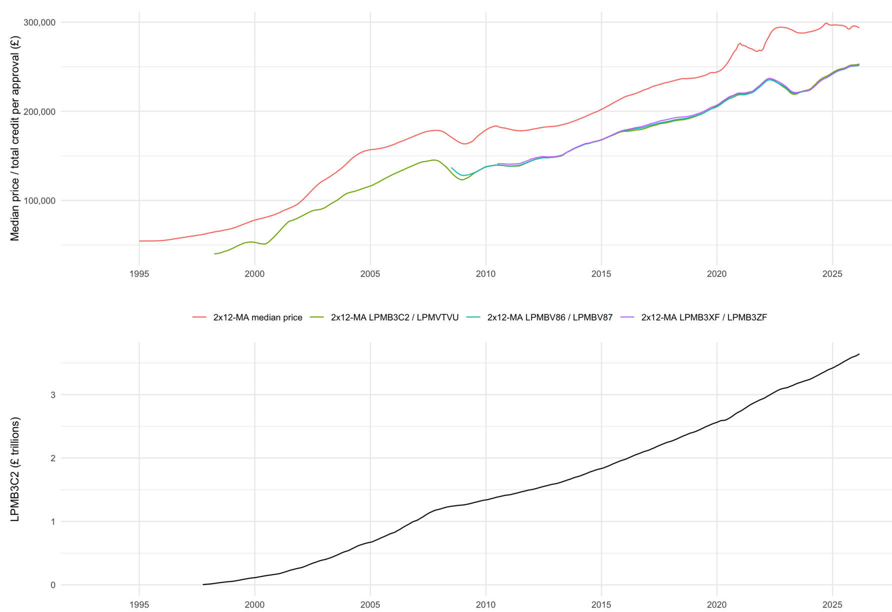

Figure A24: Moving average median price and moving average total credit per approval, and LPMB3C2 cumulative credit.

The LPMVTVU series appears to overcount approvals relative to completed transactions: in Figure A25 it exceeds Land Registry transactions until August 2007. If the early series overcounts, this implies that the LPMB3C2 / LPMVTVU line in Figure A24 above should be closer to the median price, which would represent more generous lending standards with lower equity requirements.

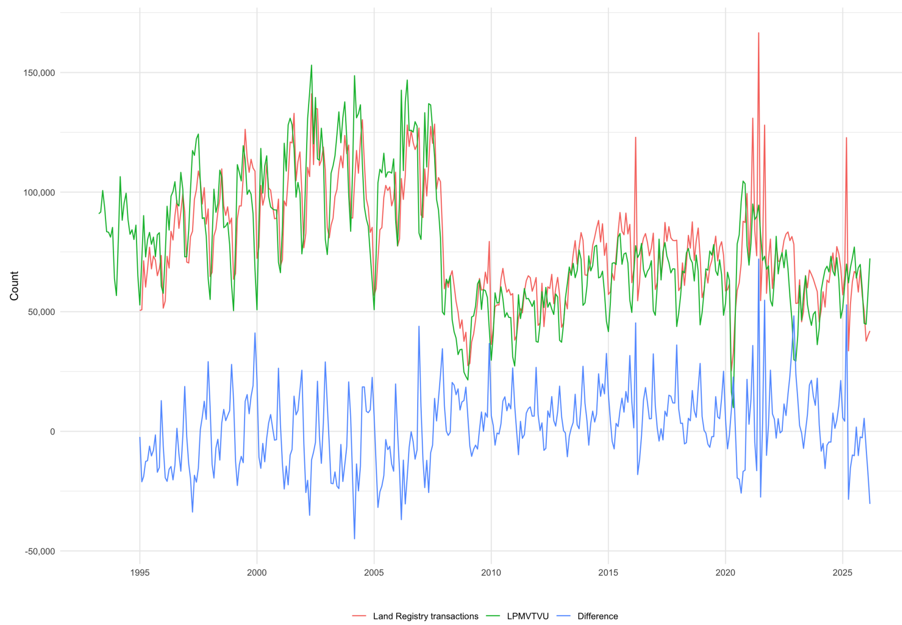

Figure A25: Land Registry transactions versus number of credit approvals.

In Figure A26, two alternate presentations of the median price versus total credit are shown.

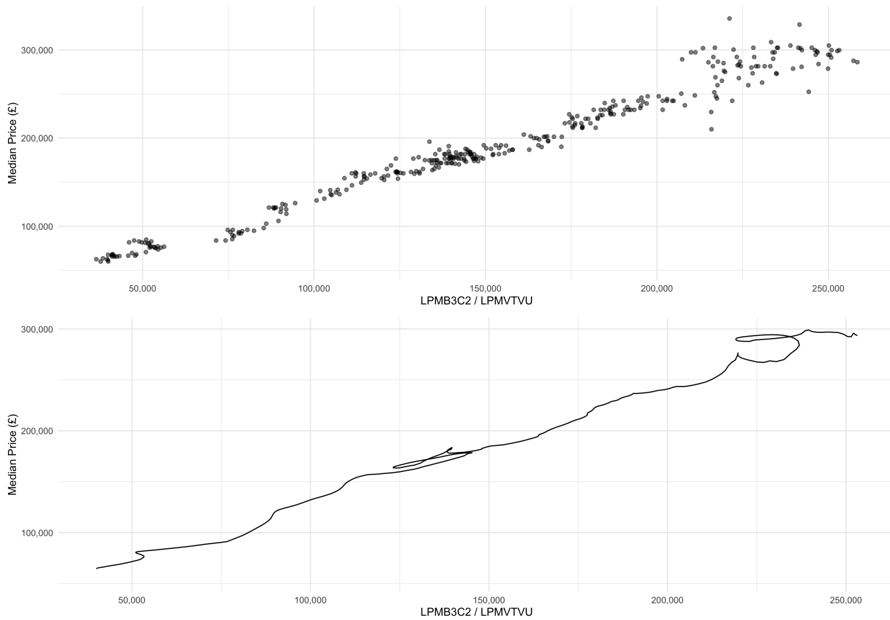

Figure A26: Median price versus total credit per approval, and moving average median price versus moving average total credit per approval.

In Figure A27, the monthly and yearly delta of both median house prices and credit is plotted. We note that credit leads prices except during the previous crash and recovery when they synchronised.

Figure A27: Median price monthly delta versus credit per approval monthly delta, and median price yearly delta versus credit per approval yearly delta.

A.4.6 Tail Transactions

Figure A28 shows the number of tail transactions at different quantiles.

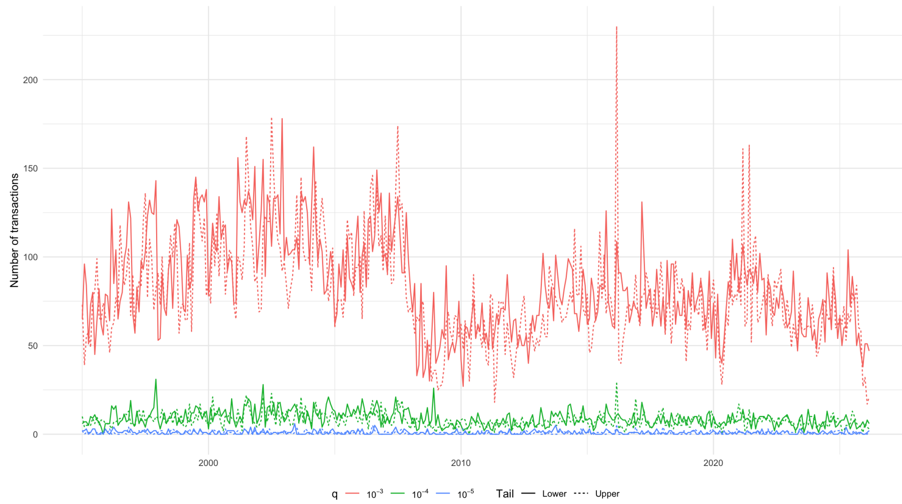

Figure A28: Tail transactions over time using a 2×12 moving average window, by quantiles, lower and upper.

A.4.7 Alternative Probability Density Functions

In Figure 1, we observe symmetric boundary-degenerate propagation in quantile space, hence $v(q,t) \propto q(1-q)$ to first order.

Table A1 compares candidate distributions on this criterion. Each velocity factor is stated in the log-price coordinate in which the model operates, as the reciprocal Jacobian $dq/d\ell$ with scale constants omitted.

Table A1: Boundary behaviour of candidate distributions in the log-price coordinate

Distribution	**Velocity factor (reciprocal Jacobian $dq/d\ell$)**	**Boundary behaviour**
Logistic / Log-logistic	$q(1-q)$ (exact at all q)	Symmetric linear vanishing at both boundaries; exact, with a single scale
Normal / Lognormal	$\phi(\Phi^{-1}(q)) \approx q\sqrt{2\ln(1/q)}$ as $q \to 0$	Symmetric; exceeds linear near the boundaries, so the tails are traversed faster and residence there is shorter
Pareto	$(1-q)$ (exact at all q)	Asymmetric; linear vanishing at the top, no vanishing at the bottom
Double Pareto lognormal	q lower tail, $(1-q)$ upper tail (asymptotic)	Linear vanishing at both boundaries to first order, with tail rates β and α; symmetric iff $\alpha = \beta$

The boundary criterion excludes the Gaussian, whose velocity factor exceeds linear near the boundaries so that the tails are traversed too quickly to produce the observed extended residence, and the Pareto, whose vanishing is one-sided. It does not, by itself, exclude a double Pareto lognormal with $\alpha = \beta$, whose first-order boundary behaviour coincides with the logistic.

That alternative is separated by the global structure rather than the tails: in the logistic family the relation $dq/d\ell = q(1-q)/s$ holds exactly at every quantile with a single scale, the linearity of log-price in log-odds across the entire distribution (Figures 4 and A21), whereas in the double Pareto lognormal it holds only asymptotically, tail by tail, and with two rates.

Additionally, the Cullen-Frey coordinates (Figure A4) place the distributions in the logistic / log-logistic region with kurtosis near 4.2, distinct from the normal distribution kurtosis of 3, providing direct empirical support for the logistic family over Gaussian alternatives.

The log-logistic form is not merely a convenient empirical fit. The log-odds coordinate

$$z = \log\left(\frac{q}{1-q}\right),$$

linearises both the growth rate and the capital density into the same uniform advection equation (3.4) because the logistic density

$$f(z) = q(1-q),$$

is simultaneously the Jacobian of the coordinate transformation and the velocity factor in the quantile-space dynamics since

$$\frac{dz}{dq} = \frac{1}{q(1-q)}, \qquad \frac{dq}{dz} = q(1-q).$$

This analytical closure occurs exactly within the logistic family and approximately within the observed empirical regime. For other distributional forms, the coordinate that linearises one quantity would not linearise the other.

A.4.8 Accumulation at the Margin

As the ridge peak approaches the margin, its leading-edge velocity vanishes while the trailing flux continues to deliver capital into the lower tail at the rate set by the wave's interior speed. The continuity equation converts this imbalance into accumulation. Near the boundary $(1-q) \to 1$, so the flux reduces to

$$J = vf \approx -\kappa q f,$$

and its divergence is

$$\partial_q J \approx -\kappa \partial_q (qf) = -\kappa\big(f + q\partial_q f\big) \xrightarrow[q\to 0]{} -\kappa f,$$

the term $q\partial_q f$ vanishing provided f approaches the boundary with bounded slope. This holds throughout the approach: the density rises smoothly as capital arrives, and the degenerate profile with unbounded slope arises only in the $q \to 0$ limit, which the finite-time argument of Section 4.1 does not require. Since $f > 0$,

$$\partial_t f = -\partial_q J \approx \kappa f > 0 \qquad \text{near } q = 0,$$

so capital density accumulates in a neighbourhood of the margin throughout the approach. The kinematics establish this accumulation but cannot reverse its sign; the reversal is the price correction, and it requires the behavioural premise of Section 4.

A.4.9 Repeat Sales Analysis

In Figure A29 we plot a repeat sales analysis.

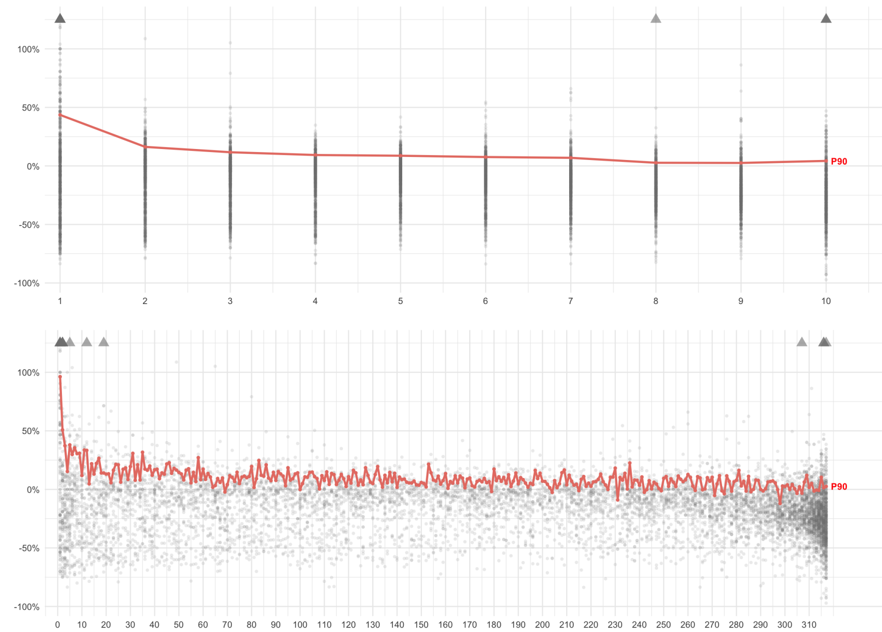

Figure A29: Repeat sales analysis: maximum intermediate price as a percentage versus decile (top) or quantile (bottom), with a 90th percentile overlay. Arrows indicate outliers.

For every property that sold both at some point between 2003 and 2007, and in 2009, we pair each pre-2008 sale with that property's 2009 sale. We then keep only pairs where the property's quantile rank shifted by at most 1 between the two sales, i.e. the property held roughly the same position in the national price distribution thereby filtering for like-for-like sales without major refurbishment or decline. For each surviving pair, we find the highest price the property fetched in the specified window, and calculate a percentage above the 2009 price.

We hypothesise that properties at the margin revert to their rebuild cost (zero land value) in the trough of a crash but lack the data to test this claim.

A.5 Null Models and Simulations

We test whether the observed monotonic propagation of growth rank leadership can arise from mechanisms that do not involve a travelling wave. Five null models are simulated, each over 30 years with log-prices initialised at empirical cross-sectional medians and projected onto the log-logistic manifold at each step.

Table A2 describes the five null models tested.

Table A2: Null models

Model	**Mechanism**
Base	Independent Ornstein-Uhlenbeck processes with spatially correlated noise
Combined	Asymmetric mean reversion, 18-year boom-bust cycle, leverage effect
Diffusion	Symmetric spatial spreading via discrete Laplacian
Spatial Autoregressive	Each quantile pulled toward lagged return of its more expensive neighbour; coupling strength β swept across 0.02–0.50
Credit-leverage	Common AR(1) credit cycle with price-dependent sensitivity

The model outputs are shown in Figure A30. Sensitivity to correlation length and coupling strength is shown in Figures A31 and A32 respectively.

None of the null models reproduce the observed travelling wave pattern. There is no extended time spent in the tails and no clean ordered propagation.

The closest is the Spatial Autoregressive model at higher coupling strengths. At $\beta \geq 0.2$ the model produces monotonic propagation, but with a much shorter period than the empirical cycle and a non-logistic trajectory shape (Figure A32); at intermediate strengths leadership oscillates or reverses. The distinction with the empirical regularity is therefore the logistic trajectory and the cycle time, not the presence of directional propagation.

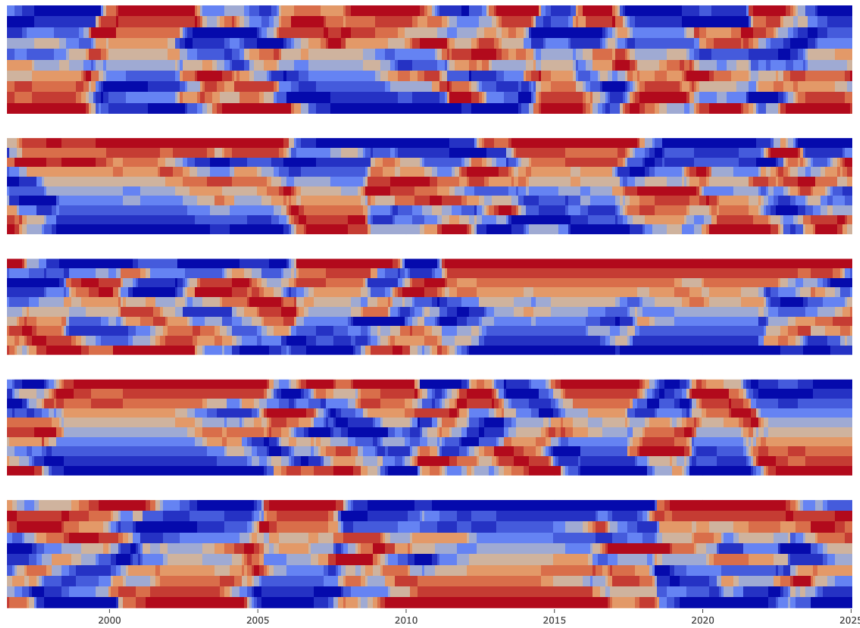

Figure A30: Null models by type: Base, Combined, Diffusion, Spatial Autoregressive, Credit-leverage; correlation length of 15 months.

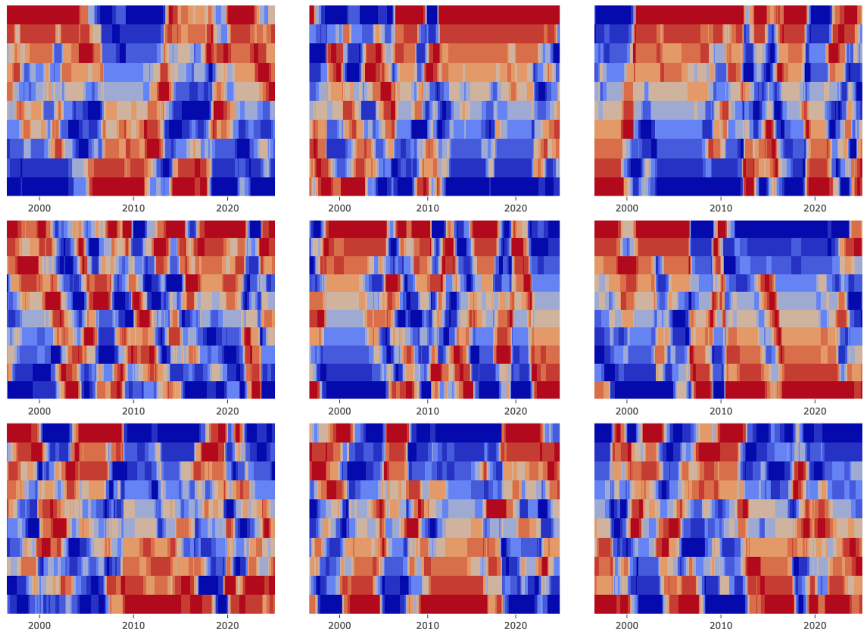

Figure A31: Correlation length sensitivity analysis. Rows: Diffusion, Spatial Autoregressive, Credit-leverage null models; Columns: 5, 15, 30 months.

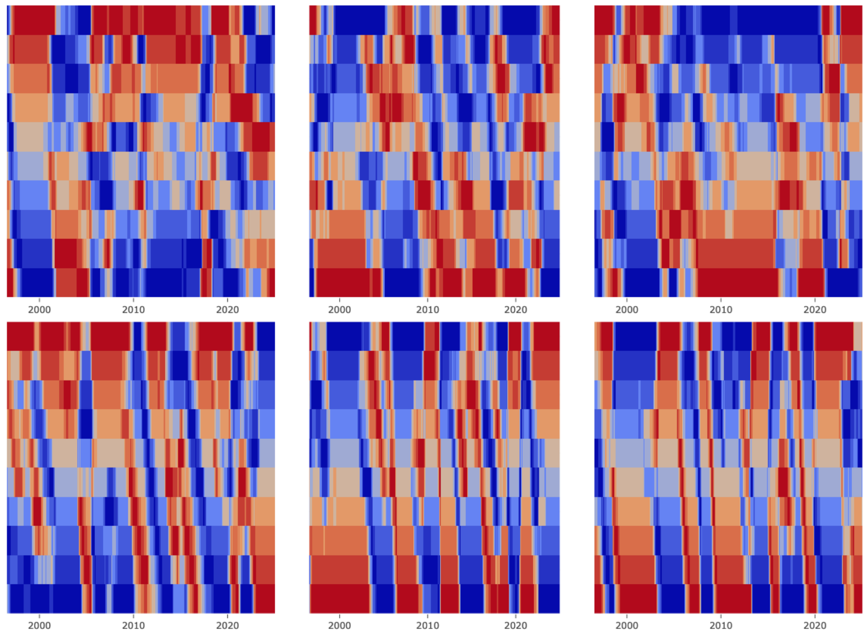

Figure A32: Coupling strength sensitivity analysis for the Spatial Autoregressive model; Beta of 0.02, 0.05, 0.1, 0.2, 0.35, 0.5.

B Neal Hudson's Heat Map

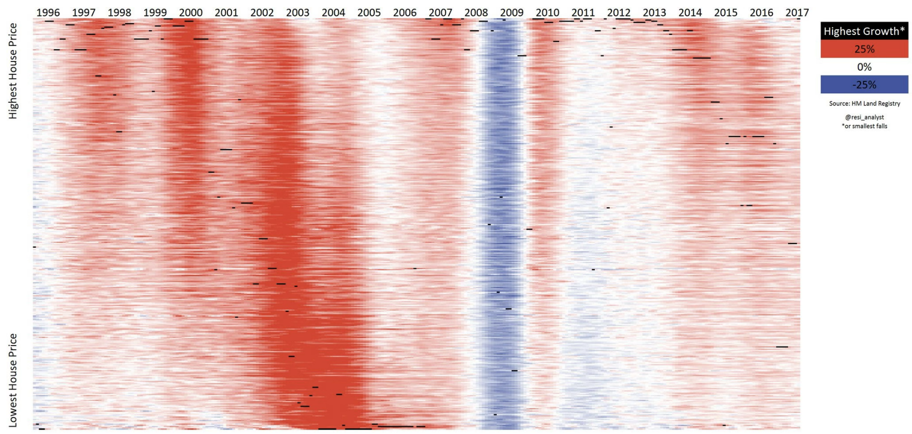

Figure B1: Annual change in Local Authority house prices, by Neal Hudson of Built Place `https://builtplace.com`.

Ordering of Local Authorities was based on the latest price; growth was based on a three-month rolling average of monthly prices. The growth did not use a log.

The overlay of black dots shows the single fastest-growing Local Authority at any point in time.

C About the Author

I graduated with an MBA in 2008 into the great financial crisis. This experience led me to want to understand the root cause of the housing market crash. In subsequent years, I project managed a medium-sized London property project through planning; initially 14 units, later 27 units. This was a perfect education in the role of land in the economy. https://uk.linkedin.com/in/chris-beech-0

D Code Availability

This paper, all code used to produce the results, and data availability, is listed at my GitHub repository[1].

E AI Usage Statement

An LLM (primarily Anthropic's Claude) under the author's direction was used to:

- Provide iterative editorial feedback.
- Provide supporting mathematical derivations and consistency checks (cf. Wolfram Alpha), with all results verified by the author.
- Suggest the similarity and reference to Lighthill and Whitham[14].
- Help write most of the complex functions in R.
- Integrate almost all of the R code into the 'Main analysis' script, enabling composition of all plots, particularly the multi-panel figures, from a single command.
- Write the R model simulation, based on the paper's framework.
- Suggest the null models (an addition that would not otherwise have been included) and to write the corresponding R code.

F Funding Statement

This work was done in my spare time with no grant support.

G Competing Interests

I have no competing interests.

H Acknowledgements

This work would not have existed had I not read Martin Wolf's *Financial Times* column on the land cycle[36], which advocated a land value tax and referenced the work of Fred Harrison. I am grateful to Neal Hudson for allowing the inclusion of his plot, and especially so to Bin Chi for making public the code for the price per square meter dataset. Furthermore, I thank the significant contributions of the R development community for the software tools essential to this research. Finally, I thank those friends who have listened to these ideas over the years.

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